CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

FRANK COELHO & SONS DAIRY LP

2031 E WASHINGTON RD

EL NIDO CA 95317

NOTES:

PREPARED BY:

CARDOSO AG SERVICES

MARIANN CARDOSO
PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Frank Coelho & Sons, LP

Physical address of dairy:

Number and Street	City	County	Zip Code
2031 Washington RD	El Nido	Merced	95317

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1972

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X075-X070-X006-XXXX

B. OPERATORS

Coelho, Mark

Operator name: Coelho, Mark
Telephone no.: (209) 769-7911 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12775 Anchor ST	El Nido	CA	95317

Coelho, Paul

Operator name: Coelho, Paul
Telephone no.: (209) 617-6455 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12775 Anchor ST	El Nido	CA	95317

Frank Coelho & Sons LP

Operator name: Frank Coelho & Sons LP
Telephone no.: (209) 617-6455 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12775 Anchor St	El Nido	CA	95317

This operator is responsible for paying permit fees.

C. OWNERS

Coelho, Mark

Legal owner name:	Coelho, Mark	Telephone no.:	(209) 769-7911	
			Landline	Cellular
12775 Anchor ST	El Nido		CA	95317
Mailing Address Number and Street	City		State	Zip Code

Coelho, Paul

Legal owner name:	Coelho, Paul	Telephone no.:	(209) 617-6455	
			Landline	Cellular
12775 Anchor ST	El Nido		CA	95317
Mailing Address Number and Street	City		State	Zip Code

Frank Coelho & Sons LP

Legal owner name:	Frank Coelho & Sons LP	Telephone no.:		(209) 617-6455
			Landline	Cellular
12775 Anchor St	El Nido		CA	95317
Mailing Address Number and Street	City		State	Zip Code

This owner is responsible for paying permit fees.

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	715	780	200	0
Number under roof	1,968	309	0	0	0	0
Maximum number	1,968	309	715	780	200	0
Average number	1,968	309	715	780	200	0
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 83 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 74,059.66 *tons per reporting period*

Total nitrogen from manure: 907,773.66 *lbs per reporting period*

After ammonia losses (30% loss applied): 635,441.56 *lbs per reporting period*

Total phosphorus from manure: 151,133.39 *lbs per reporting period*

Total potassium from manure: 393,698.85 *lbs per reporting period*

Total salt from manure: 997,687.35 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 40,920,000 *gallons*

Total nitrogen generated: 175,146.72 *lbs*

Total phosphorus generated: 23,940.03 *lbs*

Total potassium generated: 215,104.16 *lbs*

Total salt generated: 1,105,608.49 *lbs*

	40,920,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	40,920,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
10	103	103	2	manure	X075-X080-X028-XXXX
11	33	33	2	manure	X075-X080-X028-XXXX
11a	9	9	2	manure	X075-X080-X022-XXXX
3	22	22	2	process wastewater	X075-X075-X006-XXXX
4	40	40	2	both	X075-X075-X013-XXXX
5	55	55	8	both	X075-X075-X006-XXXX
6	80	80	2	both	X075-X075-X012-XXXX X075-X075-X015-XXXX
7	78	78	2	process wastewater	X075-X075-X006-XXXX
9	78	78	2	process wastewater	X075-X075-X011-XXXX
B1	117	117	8	manure	X020-X090-X002-XXXX X075-X090-X002-XXXX
B2	35	35	1	none	X075-X090-X002-XXXX
G1	34	34	2	both	X075-X075-X007-XXXX
G2	40	40	2	both	X075-X076-X001-XXXX
P1	103	103	0	none	X074-X130-X009-XXXX
P2	103	103	2	manure	X074-X130-X009-XXXX
P3	65	65	0	none	X074-X130-X009-XXXX
P4	78	78	0	none	X074-X130-X009-XXXX
Totals for areas that were used for application	792	792	38		
Totals for areas that were not used for application	281	281	1		
Land application area totals	1,073	1,073	39		

B. CROPS AND HARVESTS

10

Field name: 10

10

10/26/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 94 Plant date: 10/26/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	1,310.70 *ton*	Dry-weight		59.2	12,000.00	2,100.00	13,400.00		8.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	13.94	136.54	23.89	152.47	910.24

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 94 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/25/2023	2,559.60 *ton*	Dry-weight		68.5	16,050.00	2,750.00	12,550.00		6.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.23	275.33	47.18	215.29	1,097.91

11

Field name: 11

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 33 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	570.90 *ton*	Dry-weight		66.3	10,400.00	2,000.00	14,400.00		9.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	17.30	121.27	23.32	167.91	1,148.53

11

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 33 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	988.50 *ton*	Dry-weight		66.4	11,400.00	2,400.00	9,100.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	29.95	229.48	48.31	183.18	925.95

11a

Field name: 11a

10/27/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 9 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	139.60 *ton*	Dry-weight		66.3	10,400.00	2,000.00	14,400.00		9.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	15.51	108.73	20.91	150.54	1,029.77

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 9 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	239.40 *ton*	Dry-weight		66.5	11,400.00	2,400.00	9,100.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.60	203.17	42.77	162.18	819.81

3

Field name: 3

11/02/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 14 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	144.00 *ton*	Dry-weight		44.0	14,900.00	2,500.00	17,400.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.29	171.65	28.80	200.45	783.36

05/30/2023: Corn, silage

Crop: Corn, silage Acres planted: 14 Plant date: 05/30/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	335.50 *ton*	Dry-weight		69.2	12,500.00	1,900.00	16,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	23.96	184.53	28.05	246.53	915.24

4

Field name: 4

10/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 37 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	572.20 *ton*	Dry-weight		53.5	13,400.00	2,000.00	17,300.00		6.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	15.46	192.72	28.76	248.81	934.85

4

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	1,190.30 *ton*	Dry-weight		69.2	12,500.00	1,900.00	16,700.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	32.17	247.71	37.65	330.94	1,228.65

5

Field name: 5

11/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 55 Plant date: 11/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	177.00 *ton*	Dry-weight		57.4	39,700.00	3,700.00	29,700.00		12.90
05/17/2023	168.00 *ton*	Dry-weight		57.4	39,700.00	3,700.00	29,700.00		12.90
06/15/2023	154.00 *ton*	Dry-weight		57.4	39,700.00	3,700.00	29,700.00		12.90
07/22/2023	61.00 *ton*	Dry-weight		33.7	30,865.00	3,065.00	27,065.00		10.67
08/11/2023	77.00 *ton*	Dry-weight		33.7	30,865.00	3,065.00	27,065.00		10.67
09/11/2023	70.00 *ton*	Dry-weight		47.4	26,000.00	2,800.00	27,400.00		9.55
10/18/2023	65.00 *ton*	Dry-weight		47.4	26,000.00	2,800.00	27,400.00		9.50
11/09/2023	50.00 *ton*	Dry-weight		47.4	26,000.00	2,800.00	27,400.00		9.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.95	501.57	48.71	416.58	1,689.47

6

Field name: 6

6

10/18/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 72 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	1,025.00 *ton*	Dry-weight		62.4	16,100.00	2,600.00	21,800.00		9.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	14.24	172.36	27.83	233.38	968.85

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,969.50 *ton*	Dry-weight		65.1	12,850.00	2,100.00	15,850.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.35	245.35	40.10	302.63	1,183.78

7

Field name: 7

10/19/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 75 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,028.30 *ton*	Dry-weight		62.4	17,700.00	2,900.00	23,500.00		9.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	13.71	182.49	29.90	242.29	974.33

7

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	2,024.30 *ton*	Dry-weight		65.9	13,500.00	2,300.00	17,600.00		6.85

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.99	248.50	42.34	323.97	1,260.92

9

Field name: 9

10/17/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 72 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/30/2023	1,309.20 *ton*	Dry-weight		58.0	17,600.00	2,700.00	21,900.00		9.05

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	18.18	268.82	41.24	334.50	1,382.30

05/20/2023: Corn, silage

Crop: Corn, silage Acres planted: 72 Plant date: 05/20/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	2,042.80 *ton*	Dry-weight		63.0	12,750.00	2,350.00	14,650.00		5.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	28.37	267.69	49.34	307.58	1,165.25

B1

Field name: B1

11/09/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 117 Plant date: 11/09/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/18/2023	376.00 *ton*	Dry-weight		73.3	42,600.00	4,400.00	34,200.00		14.35
05/17/2023	362.00 *ton*	Dry-weight		73.3	42,600.00	4,400.00	34,200.00		14.35
06/15/2023	328.00 *ton*	Dry-weight		73.3	42,600.00	4,400.00	34,200.00		14.35
07/22/2023	129.00 *ton*	Dry-weight		23.6	34,320.00	3,333.00	30,790.00		13.35
08/11/2023	164.00 *ton*	Dry-weight		23.6	34,320.00	3,333.00	30,790.00		13.35
09/11/2023	150.00 *ton*	Dry-weight		37.1	30,300.00	2,750.00	30,800.00		12.35
10/18/2023	140.00 *ton*	Dry-weight		37.1	30,300.00	2,750.00	30,800.00		12.35
11/09/2023	105.00 *ton*	Dry-weight		37.1	30,300.00	2,750.00	30,800.00		12.35

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	14.99	467.28	45.84	415.02	1,733.53

B2

Field name: B2

10/19/2022: Pistachios

Crop: Pistachios Acres planted: 35 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	176.10 *ton*	Dry-weight		55.8	12,500.00	1,300.00	7,900.00		1.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	390.00	60.00	300.00	3.00
Total actual harvest content	5.03	55.60	5.78	35.14	44.48

G1

Field name: G1

G1

10/19/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 34 Plant date: 10/19/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	685.60 *ton*	Dry-weight		61.6	11,800.00	2,300.00	16,400.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	20.16	182.74	35.62	253.98	1,068.57

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,054.60 *ton*	Dry-weight		65.7	11,400.00	2,600.00	17,600.00		7.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	31.02	242.57	55.32	374.49	1,574.58

G2

Field name: G2

10/20/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 38 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	415.90 *ton*	Dry-weight		58.4	13,800.00	2,200.00	18,400.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.94	125.66	20.03	167.55	810.44

G2

05/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 38 Plant date: 05/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/08/2023	1,375.80 *ton*	Dry-weight		70.8	11,200.00	2,500.00	16,200.00		6.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	36.21	236.81	52.86	342.53	1,310.92

P2

Field name: P2

10/22/2022: Wheat, silage, soft dough

Crop: Wheat, silage, soft dough Acres planted: 70 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/20/2023	737.10 *ton*	Dry-weight		59.0	13,400.00	2,500.00	16,900.00		12.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	18.00	198.00	30.60	149.40	0.00
Total actual harvest content	10.53	115.70	21.59	145.92	1,079.33

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 95 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/09/2023	1,856.90 *ton*	Dry-weight		66.9	14,800.00	2,550.00	12,150.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	19.55	191.51	33.00	157.22	918.72

NUTRIENT BUDGET

A. LAND APPLICATIONS

10 - 10/26/2022: Wheat, silage, soft dough

Field name: 10

Crop: Wheat, silage, soft dough — Plant date: 10/26/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	182.24	23.59	36.57	0.00	385.00 *ton*
Application event totals		182.24	23.59	36.57	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.18	0.00	0.00	0.00	10,080,000.00 *gal*
Application event totals		0.18	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.17	0.00	0.00	0.00	9,360,000.00 *gal*
Application event totals		0.17	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.16	0.00	0.00	0.00	9,273,600.00 *gal*
Application event totals		0.16	0.00	0.00	0.00	

10 - 06/12/2023: Corn, silage

Field name: 10

Crop: Corn, silage — Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

10 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	284.27	154.15	94.76	7,354.13	2,300.00 *ton*
Application event totals		284.27	154.15	94.76	7,354.13	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.24	0.00	0.00	0.00	13,680,000.00 *gal*
Canal	Surface water	0.53	0.00	0.00	14.93	6,005,489.00 *gal*
Application event totals		0.78	0.00	0.00	14.93	

07/15/2023	Shank	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.24	0.00	0.00	0.00	13,680,000.00 *gal*
Canal	Surface water	0.92	0.00	0.00	25.79	10,375,718.00 *gal*
Application event totals		1.16	0.00	0.00	25.79	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.22	0.00	0.00	0.00	12,240,000.00 *gal*
Canal	Surface water	0.85	0.00	0.00	23.88	9,608,745.00 *gal*
Application event totals		1.07	0.00	0.00	23.88	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.23	0.00	0.00	0.00	12,960,000.00 *gal*
Canal	Surface water	0.63	0.00	0.00	17.74	7,137,862.00 *gal*
Application event totals		0.86	0.00	0.00	17.74	

10 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.20	0.00	0.00	0.00	11,520,000.00 *gal*
Canal	Surface water	0.77	0.00	0.00	21.61	8,693,583.00 *gal*
Application event totals		0.98	0.00	0.00	21.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.20	0.00	0.00	0.00	11,520,000.00 *gal*
Canal	Surface water	0.96	0.00	0.00	26.84	10,798,327.00 *gal*
Application event totals		1.16	0.00	0.00	26.84	

11 - 10/27/2022: Wheat, silage, soft dough

Field name: 11

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	161.80	20.95	32.47	0.00	120.00 *ton*
Application event totals		161.80	20.95	32.47	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.27	0.00	0.00	0.00	5,400,000.00 *gal*
Application event totals		0.27	0.00	0.00	0.00	

11 - 10/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.25	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		0.25	0.00	0.00	0.00	

11 - 05/20/2023: Corn, silage

Field name: 11

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	281.64	152.73	93.88	7,286.30	800.00 *ton*
Application event totals		281.64	152.73	93.88	7,286.30	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.94	0.00	0.00	54.38	7,680,000.00 *gal*
Application event totals		1.94	0.00	0.00	54.38	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.38	0.00	0.00	0.00	7,440,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.41	0.00	0.00	0.00	8,040,000.00 *gal*
Application event totals		0.41	0.00	0.00	0.00	

11 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.41	0.00	0.00	0.00	8,160,000.00 *gal*
Application event totals		0.41	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.38	0.00	0.00	0.00	7,440,000.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.39	0.00	0.00	0.00	7,800,000.00 *gal*
Application event totals		0.39	0.00	0.00	0.00	

11a - 10/27/2022: Wheat, silage, soft dough

Field name: 11a

Crop: Wheat, silage, soft dough

Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/11/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	143.38	18.56	28.77	0.00	29.00 *ton*
Application event totals		143.38	18.56	28.77	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/21/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.27	0.00	0.00	0.00	1,471,390.00 *gal*
Application event totals		0.27	0.00	0.00	0.00	

11a - 10/27/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.25	0.00	0.00	0.00	1,373,297.00 *gal*
Application event totals		0.25	0.00	0.00	0.00	

11a - 05/26/2023: Corn, silage

Field name: 11a

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	271.08	147.00	90.36	7,013.07	210.00 *ton*
Application event totals		271.08	147.00	90.36	7,013.07	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.39	0.00	0.00	0.00	2,092,643.00 *gal*
Application event totals		0.39	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.38	0.00	0.00	0.00	2,027,248.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.41	0.00	0.00	0.00	2,190,736.00 *gal*
Application event totals		0.41	0.00	0.00	0.00	

11a - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.41	0.00	0.00	0.00	2,223,433.00 *gal*
Application event totals		0.41	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.38	0.00	0.00	0.00	2,027,248.00 *gal*
Application event totals		0.38	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.39	0.00	0.00	0.00	2,125,341.00 *gal*
Application event totals		0.39	0.00	0.00	0.00	

3 - 11/02/2022: Wheat, silage, soft dough

Field name: 3

Crop: Wheat, silage, soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.45	9.60	96.82	502.13	216,000.00 *gal*
Canal	Surface water	1.22	0.00	0.00	34.04	2,039,325.00 *gal*
Application event totals		85.66	9.60	96.82	536.17	

3 - 11/02/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.08	9.65	96.02	518.44	192,000.00 *gal*
Canal	Surface water	1.11	0.00	0.00	31.16	1,866,733.00 *gal*
Application event totals		77.20	9.65	96.02	549.59	

03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	66.57	8.44	84.02	453.63	168,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,269,231.00 *gal*
Application event totals		66.57	8.44	84.02	453.63	

3 - 05/30/2023: Corn, silage

Field name: 3

Crop: Corn, silage

Plant date: 05/30/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,650,000.00 *gal*
Canal	Surface water	0.68	0.00	0.00	19.13	1,145,999.00 *gal*
Application event totals		0.68	0.00	0.00	19.13	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	57.06	7.24	72.01	388.83	144,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,776,923.00 *gal*
Canal	Surface water	0.69	0.00	0.00	19.41	1,162,834.00 *gal*
Application event totals		57.76	7.24	72.01	408.24	

3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.72	12.42	127.98	628.88	336,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,269,231.00 *gal*
Canal	Surface water	1.20	0.00	0.00	33.73	2,020,917.00 *gal*
Application event totals		99.92	12.42	127.98	662.61	

07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.62	10.64	109.70	539.04	288,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,523,077.00 *gal*
Canal	Surface water	0.76	0.00	0.00	21.31	1,277,056.00 *gal*
Application event totals		85.38	10.64	109.70	560.35	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,269,231.00 *gal*
Canal	Surface water	0.48	0.00	0.00	13.40	802,860.00 *gal*
Application event totals		0.48	0.00	0.00	13.40	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,142,308.00 *gal*
Canal	Surface water	0.70	0.00	0.00	19.71	1,181,241.00 *gal*
Application event totals		0.70	0.00	0.00	19.71	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,269,231.00 *gal*
Canal	Surface water	0.60	0.00	0.00	16.78	1,005,503.00 *gal*
Application event totals		0.60	0.00	0.00	16.78	

3 - 05/30/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	1,396,154.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.26	734,579.00 *gal*
Application event totals		0.44	0.00	0.00	12.26	

4 - 10/18/2022: Wheat, silage, soft dough

Field name: 4

Crop: Wheat, silage, soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.21	9.69	97.69	506.65	576,000.00 *gal*
Canal	Surface water	1.22	0.00	0.00	34.04	5,389,644.00 *gal*
Application event totals		86.42	9.69	97.69	540.69	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	89.96	11.41	113.54	613.02	600,000.00 *gal*
Canal	Surface water	1.11	0.00	0.00	31.16	4,933,508.00 *gal*
Application event totals		91.08	11.41	113.54	644.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	82.77	10.50	104.45	563.98	552,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,300,000.00 *gal*
Application event totals		82.77	10.50	104.45	563.98	

4 - 05/26/2023: Corn, silage

4 - 05/26/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	150.72	81.73	50.24	3,899.16	480.00 *ton*
Application event totals		150.72	81.73	50.24	3,899.16	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,290,000.00 *gal*
Canal	Surface water	0.68	0.00	0.00	19.13	3,028,712.00 *gal*
Application event totals		0.68	0.00	0.00	19.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	4,620,000.00 *gal*
Canal	Surface water	0.69	0.00	0.00	19.41	3,073,203.00 *gal*
Application event totals		0.69	0.00	0.00	19.41	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,300,000.00 *gal*
Canal	Surface water	1.20	0.00	0.00	33.73	5,340,995.00 *gal*
Application event totals		1.20	0.00	0.00	33.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.05	12.08	124.52	611.88	864,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,960,000.00 *gal*
Canal	Surface water	0.76	0.00	0.00	21.31	3,375,077.00 *gal*
Application event totals		96.81	12.08	124.52	633.20	

4 - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	85.38	10.74	110.68	543.90	768,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,300,000.00 *gal*
Canal	Surface water	0.48	0.00	0.00	13.40	2,121,845.00 *gal*
Application event totals		85.86	10.74	110.68	557.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	2,970,000.00 *gal*
Canal	Surface water	0.70	0.00	0.00	19.71	3,121,852.00 *gal*
Application event totals		0.70	0.00	0.00	19.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,300,000.00 *gal*
Canal	Surface water	0.60	0.00	0.00	16.78	2,657,400.00 *gal*
Application event totals		0.60	0.00	0.00	16.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	3,630,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.26	1,941,386.00 *gal*
Application event totals		0.44	0.00	0.00	12.26	

5 - 11/01/2019: Alfalfa, hay

Field name: 5

Crop: Alfalfa, hay

Plant date: 11/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

5 - 11/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.60	21.85	56.34	170.71	384,000.00 *gal*
Canal	Surface water	2.15	0.00	0.00	60.10	14,147,319.00 *gal*
Application event totals		122.75	21.85	56.34	230.81	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	120.60	21.85	56.34	170.71	384,000.00 *gal*
Canal	Surface water	1.80	0.00	0.00	50.33	11,846,465.00 *gal*
Application event totals		122.40	21.85	56.34	221.04	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.91	0.00	0.00	53.61	12,618,848.00 *gal*
Application event totals		1.91	0.00	0.00	53.61	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.03	0.00	0.00	28.98	6,821,087.00 *gal*
Application event totals		1.03	0.00	0.00	28.98	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.06	0.00	0.00	29.59	6,964,483.00 *gal*
Application event totals		1.06	0.00	0.00	29.59	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	32.68	7,691,240.00 *gal*
Application event totals		1.17	0.00	0.00	32.68	

5 - 11/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.71	0.00	0.00	47.95	11,285,917.00 *gal*
Application event totals		1.71	0.00	0.00	47.95	

09/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	278.57	34.36	66.27	0.00	1,430.00 *ton*
Application event totals		278.57	34.36	66.27	0.00	

6 - 10/18/2022: Wheat, silage, soft dough

Field name: 6

Crop: Wheat, silage, soft dough

Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.22	0.00	0.00	34.04	10,487,956.00 *gal*
Application event totals		1.22	0.00	0.00	34.04	

01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.16	12.19	121.36	655.25	1,248,000.00 *gal*
Canal	Surface water	1.11	0.00	0.00	31.16	9,600,340.00 *gal*
Application event totals		97.27	12.19	121.36	686.41	

6 - 10/18/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.86	12.66	126.03	680.45	1,296,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Application event totals		99.86	12.66	126.03	680.45	

6 - 05/20/2023: Corn, silage

Field name: 6

Crop: Corn, silage Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	142.80	77.44	47.60	3,694.38	885.00 *ton*
Application event totals		142.80	77.44	47.60	3,694.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,580,000.00 *gal*
Canal	Surface water	0.68	0.00	0.00	19.13	5,893,710.00 *gal*
Application event totals		0.68	0.00	0.00	19.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.86	12.66	126.03	680.45	1,296,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	9,240,000.00 *gal*
Canal	Surface water	0.69	0.00	0.00	19.41	5,980,287.00 *gal*
Application event totals		100.55	12.66	126.03	699.86	

6 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	1.20	0.00	0.00	33.73	10,393,288.00 *gal*
Application event totals		1.20	0.00	0.00	33.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.75	11.04	113.76	559.00	1,536,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,920,000.00 *gal*
Canal	Surface water	0.76	0.00	0.00	21.31	6,567,717.00 *gal*
Application event totals		88.51	11.04	113.76	580.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	0.48	0.00	0.00	13.40	4,128,996.00 *gal*
Application event totals		0.48	0.00	0.00	13.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	5,940,000.00 *gal*
Canal	Surface water	0.70	0.00	0.00	19.71	6,074,956.00 *gal*
Application event totals		0.70	0.00	0.00	19.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	0.60	0.00	0.00	16.78	5,171,156.00 *gal*
Application event totals		0.60	0.00	0.00	16.78	

6 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,260,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.26	3,777,833.00 *gal*
Application event totals		0.44	0.00	0.00	12.26	

7 - 10/19/2022: Wheat, silage, soft dough

Field name: 7

Crop: Wheat, silage, soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.02	16.02	51.78	348.65	1,152,000.00 *gal*
Canal	Surface water	1.22	0.00	0.00	34.04	10,924,955.00 *gal*
Application event totals		52.23	16.02	51.78	382.68	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.31	11.71	116.50	629.04	1,248,000.00 *gal*
Canal	Surface water	1.11	0.00	0.00	31.16	10,000,354.00 *gal*
Application event totals		93.43	11.71	116.50	660.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.87	12.16	120.99	653.23	1,296,000.00 *gal*
Irrigation Well	Ground water	5.05	0.00	0.00	0.00	7,200,000.00 *gal*
Application event totals		100.91	12.16	120.99	653.23	

7 - 05/29/2023: Corn, silage

7 - 05/29/2023: Corn, silage

Field name: 7

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.47	0.00	0.00	0.00	7,800,000.00 *gal*
Canal	Surface water	0.68	0.00	0.00	19.13	6,139,282.00 *gal*
Application event totals		6.15	0.00	0.00	19.13	

06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.87	12.16	120.99	653.23	1,296,000.00 *gal*
Irrigation Well	Ground water	5.89	0.00	0.00	0.00	8,400,000.00 *gal*
Canal	Surface water	0.69	0.00	0.00	19.41	6,229,466.00 *gal*
Application event totals		102.45	12.16	120.99	672.64	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.31	11.71	116.50	629.04	1,248,000.00 *gal*
Irrigation Well	Ground water	5.05	0.00	0.00	0.00	7,200,000.00 *gal*
Canal	Surface water	1.20	0.00	0.00	33.73	10,826,342.00 *gal*
Application event totals		98.57	11.71	116.50	662.77	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.05	0.00	0.00	0.00	7,200,000.00 *gal*
Canal	Surface water	0.76	0.00	0.00	21.31	6,841,371.00 *gal*
Application event totals		5.81	0.00	0.00	21.31	

7 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.05	0.00	0.00	0.00	7,200,000.00 *gal*
Canal	Surface water	0.48	0.00	0.00	13.40	4,301,037.00 *gal*
Application event totals		5.53	0.00	0.00	13.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.24	10.60	109.21	536.64	1,536,000.00 *gal*
Irrigation Well	Ground water	4.54	0.00	0.00	0.00	6,480,000.00 *gal*
Canal	Surface water	0.70	0.00	0.00	19.71	6,328,079.00 *gal*
Application event totals		89.49	10.60	109.21	556.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.05	0.00	0.00	0.00	7,200,000.00 *gal*
Canal	Surface water	0.60	0.00	0.00	16.78	5,386,621.00 *gal*
Application event totals		5.65	0.00	0.00	16.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.63	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.26	3,935,243.00 *gal*
Application event totals		5.06	0.00	0.00	12.26	

9 - 10/17/2022: Wheat, silage, soft dough

Field name: 9

Crop: Wheat, silage, soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

9 - 10/17/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.58	9.96	100.41	520.73	1,152,000.00 *gal*
Canal	Surface water	1.22	0.00	0.00	34.04	10,487,956.00 *gal*
Application event totals		88.79	9.96	100.41	554.76	

01/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.01	12.43	123.69	667.85	1,272,000.00 *gal*
Canal	Surface water	1.11	0.00	0.00	31.16	9,600,340.00 *gal*
Application event totals		99.12	12.43	123.69	699.01	

03/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	99.86	12.66	126.03	680.45	1,296,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Application event totals		99.86	12.66	126.03	680.45	

9 - 05/20/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 05/20/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	8,580,000.00 *gal*
Canal	Surface water	0.68	0.00	0.00	19.13	5,893,710.00 *gal*
Application event totals		0.68	0.00	0.00	19.13	

9 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	9,240,000.00 *gal*
Canal	Surface water	0.69	0.00	0.00	19.41	5,980,287.00 *gal*
Application event totals		0.69	0.00	0.00	19.41	

06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	103.56	13.13	130.69	705.65	1,344,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	1.20	0.00	0.00	33.73	10,393,288.00 *gal*
Application event totals		104.76	13.13	130.69	739.38	

07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,920,000.00 *gal*
Canal	Surface water	0.76	0.00	0.00	21.31	6,567,717.00 *gal*
Application event totals		0.76	0.00	0.00	21.31	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	87.75	11.04	113.76	559.00	1,536,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	0.48	0.00	0.00	13.40	4,128,996.00 *gal*
Application event totals		88.23	11.04	113.76	572.40	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.23	11.73	120.87	593.94	1,632,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	5,940,000.00 *gal*
Canal	Surface water	0.70	0.00	0.00	19.71	6,074,956.00 *gal*
Application event totals		93.94	11.73	120.87	613.66	

9 - 05/20/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.78	9.66	99.54	489.13	1,344,000.00 *gal*
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	6,600,000.00 *gal*
Canal	Surface water	0.60	0.00	0.00	16.78	5,171,156.00 *gal*
Application event totals		77.38	9.66	99.54	505.91	

08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	0.00	0.00	0.00	0.00	7,260,000.00 *gal*
Canal	Surface water	0.44	0.00	0.00	12.26	3,777,833.00 *gal*
Application event totals		0.44	0.00	0.00	12.26	

B1 - 11/09/2020: Alfalfa, hay

Field name: B1

Crop: Alfalfa, hay

Plant date: 11/09/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.73	0.00	0.00	0.00	19,800,000.00 *gal*
Application event totals		10.73	0.00	0.00	0.00	

06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.92	0.00	0.00	0.00	18,300,000.00 *gal*
Application event totals		9.92	0.00	0.00	0.00	

B1 - 11/09/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.57	0.00	0.00	0.00	19,500,000.00 *gal*
Application event totals		10.57	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.25	0.00	0.00	0.00	18,900,000.00 *gal*
Application event totals		10.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.76	0.00	0.00	0.00	18,000,000.00 *gal*
Application event totals		9.76	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.73	0.00	0.00	0.00	19,800,000.00 *gal*
Application event totals		10.73	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.25	0.00	0.00	0.00	18,900,000.00 *gal*
Application event totals		10.25	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.76	0.00	0.00	0.00	18,000,000.00 *gal*
Application event totals		9.76	0.00	0.00	0.00	

B1 - 11/09/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/25/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	278.85	34.40	66.33	0.00	3,045.00 *ton*
Application event totals		278.85	34.40	66.33	0.00	

B2 - 10/19/2022: Pistachios

Field name: B2

Crop: Pistachios

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.82	0.00	0.00	0.00	5,280,000.00 *gal*
Application event totals		9.82	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	4,880,000.00 *gal*
Application event totals		9.08	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.67	0.00	0.00	0.00	5,200,000.00 *gal*
Application event totals		9.67	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Subsurface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.37	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		9.37	0.00	0.00	0.00	

B2 - 10/19/2022: Pistachios

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.93	0.00	0.00	0.00	4,800,000.00 *gal*
Application event totals		8.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.82	0.00	0.00	0.00	5,280,000.00 *gal*
Application event totals		9.82	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.37	0.00	0.00	0.00	5,040,000.00 *gal*
Application event totals		9.37	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.93	0.00	0.00	0.00	4,800,000.00 *gal*
Application event totals		8.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.93	0.00	0.00	0.00	4,800,000.00 *gal*
Application event totals		8.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.08	0.00	0.00	0.00	4,880,000.00 *gal*
Application event totals		9.08	0.00	0.00	0.00	

G1 - 10/19/2022: Wheat, silage, soft dough

G1 - 10/19/2022: Wheat, silage, soft dough

Field name: G1

Crop: Wheat, silage, soft dough

Plant date: 10/19/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.73	10.55	106.31	551.36	576,000.00 *gal*
Irrigation Well	Ground water	5.20	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		97.92	10.55	106.31	551.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.15	12.80	171.53	808.19	1,344,000.00 *gal*
Irrigation Well	Ground water	4.82	0.00	0.00	0.00	3,120,000.00 *gal*
Application event totals		79.97	12.80	171.53	808.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	14.28	4.42	22.41	117.69	1,440,000.00 *gal*
Irrigation Well	Ground water	5.20	0.00	0.00	0.00	3,360,000.00 *gal*
Application event totals		19.47	4.42	22.41	117.69	

G1 - 05/25/2023: Corn, silage

Field name: G1

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	79.96	43.36	26.65	2,068.56	234.00 *ton*
Application event totals		79.96	43.36	26.65	2,068.56	

G1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	14.28	4.42	22.41	117.69	1,440,000.00 *gal*
Irrigation Well	Ground water	7.05	0.00	0.00	0.00	4,560,000.00 *gal*
Canal	Surface water	0.53	0.00	0.00	14.93	2,172,198.00 *gal*
Application event totals		21.86	4.42	22.41	132.62	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.72	12.42	127.98	628.88	816,000.00 *gal*
Irrigation Well	Ground water	7.05	0.00	0.00	0.00	4,560,000.00 *gal*
Canal	Surface water	0.92	0.00	0.00	25.79	3,752,919.00 *gal*
Application event totals		106.69	12.42	127.98	654.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.31	0.00	0.00	0.00	4,080,000.00 *gal*
Canal	Surface water	0.85	0.00	0.00	23.88	3,475,504.00 *gal*
Application event totals		7.16	0.00	0.00	23.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	95.81	12.05	124.21	610.38	792,000.00 *gal*
Irrigation Well	Ground water	6.68	0.00	0.00	0.00	4,320,000.00 *gal*
Canal	Surface water	0.63	0.00	0.00	17.74	2,581,780.00 *gal*
Application event totals		103.13	12.05	124.21	628.13	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.94	0.00	0.00	0.00	3,840,000.00 *gal*
Canal	Surface water	0.77	0.00	0.00	21.61	3,144,487.00 *gal*
Application event totals		6.71	0.00	0.00	21.61	

G1 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.94	0.00	0.00	0.00	3,840,000.00 *gal*
Canal	Surface water	0.96	0.00	0.00	26.84	3,905,778.00 *gal*
Application event totals		6.90	0.00	0.00	26.84	

G2 - 10/20/2022: Wheat, silage, soft dough

Field name: G2

Crop: Wheat, silage, soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	76.05	8.65	87.20	452.21	528,000.00 *gal*
Irrigation Well	Ground water	5.11	0.00	0.00	0.00	3,696,000.00 *gal*
Application event totals		81.17	8.65	87.20	452.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	62.43	10.63	142.52	671.47	1,248,000.00 *gal*
Irrigation Well	Ground water	4.75	0.00	0.00	0.00	3,432,000.00 *gal*
Application event totals		67.18	10.63	142.52	671.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	11.92	3.69	18.71	98.28	1,344,000.00 *gal*
Irrigation Well	Ground water	5.11	0.00	0.00	0.00	3,696,000.00 *gal*
Application event totals		17.04	3.69	18.71	98.28	

G2 - 05/25/2023: Corn, silage

G2 - 05/25/2023: Corn, silage

Field name: G2

Crop: Corn, silage

Plant date: 05/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	83.16	45.10	27.72	2,151.38	272.00 *ton*
Application event totals		83.16	45.10	27.72	2,151.38	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	13.20	4.08	20.72	108.82	1,488,000.00 *gal*
Irrigation Well	Ground water	7.07	0.00	0.00	0.00	5,107,200.00 *gal*
Canal	Surface water	0.53	0.00	0.00	14.93	2,427,751.00 *gal*
Application event totals		20.80	4.08	20.72	123.74	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	93.52	11.76	121.24	595.78	864,000.00 *gal*
Irrigation Well	Ground water	7.07	0.00	0.00	0.00	5,107,200.00 *gal*
Canal	Surface water	0.92	0.00	0.00	25.79	4,194,439.00 *gal*
Application event totals		101.51	11.76	121.24	621.57	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	6.32	0.00	0.00	0.00	4,569,600.00 *gal*
Canal	Surface water	0.85	0.00	0.00	23.88	3,884,386.00 *gal*
Application event totals		7.18	0.00	0.00	23.88	

G2 - 05/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	90.92	11.44	117.88	579.23	840,000.00 *gal*
Irrigation Well	Ground water	6.69	0.00	0.00	0.00	4,838,400.00 *gal*
Canal	Surface water	0.63	0.00	0.00	17.74	2,885,519.00 *gal*
Application event totals		98.25	11.44	117.88	596.97	

08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.95	0.00	0.00	0.00	4,300,800.00 *gal*
Canal	Surface water	0.77	0.00	0.00	21.61	3,514,427.00 *gal*
Application event totals		6.72	0.00	0.00	21.61	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.95	0.00	0.00	0.00	4,300,800.00 *gal*
Canal	Surface water	0.96	0.00	0.00	26.84	4,365,281.00 *gal*
Application event totals		6.91	0.00	0.00	26.84	

P2 - 10/22/2022: Wheat, silage, soft dough

Field name: P2

Crop: Wheat, silage, soft dough

Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/10/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Corral solids	114.42	14.81	22.96	0.00	180.00 *ton*
Application event totals		114.42	14.81	22.96	0.00	

P2 - 10/22/2022: Wheat, silage, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/02/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	19.71	0.00	0.00	0.00	10,080,000.00 *gal*
Application event totals		19.71	0.00	0.00	0.00	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	17.60	0.00	0.00	0.00	9,000,000.00 *gal*
Application event totals		17.60	0.00	0.00	0.00	

P2 - 06/15/2023: Corn, silage

Field name: P2

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	29.82	0.00	0.00	0.00	20,700,000.00 *gal*
Application event totals		29.82	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	29.82	0.00	0.00	0.00	20,700,000.00 *gal*
Application event totals		29.82	0.00	0.00	0.00	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	28.52	0.00	0.00	0.00	19,800,000.00 *gal*
Application event totals		28.52	0.00	0.00	0.00	

P2 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	28.52	0.00	0.00	0.00	19,800,000.00 *gal*
Application event totals		28.52	0.00	0.00	0.00	

08/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	29.82	0.00	0.00	0.00	20,700,000.00 *gal*
Application event totals		29.82	0.00	0.00	0.00	

09/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	29.17	0.00	0.00	0.00	20,250,000.00 *gal*
Application event totals		29.17	0.00	0.00	0.00	

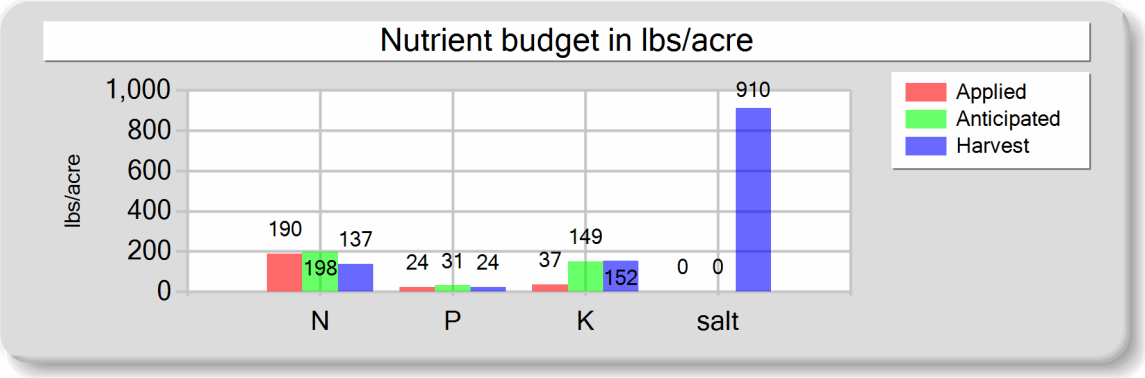
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	27.88	0.00	0.00	0.00	19,350,000.00 *gal*
Application event totals		27.88	0.00	0.00	0.00	

B. NUTRIENT BUDGET

10 - 10/26/2022: Wheat, silage, soft dough

Field name: 10 Crop: Wheat, silage, soft dough Plant date: 10/26/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	182.24	23.59	36.57	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.51	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	189.75	23.59	36.57	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	136.54	23.89	152.47	910.24
Nutrient balance	53.22	-0.30	-115.90	-910.24
Applied to removed ratio	1.39	0.99	0.24	0.00

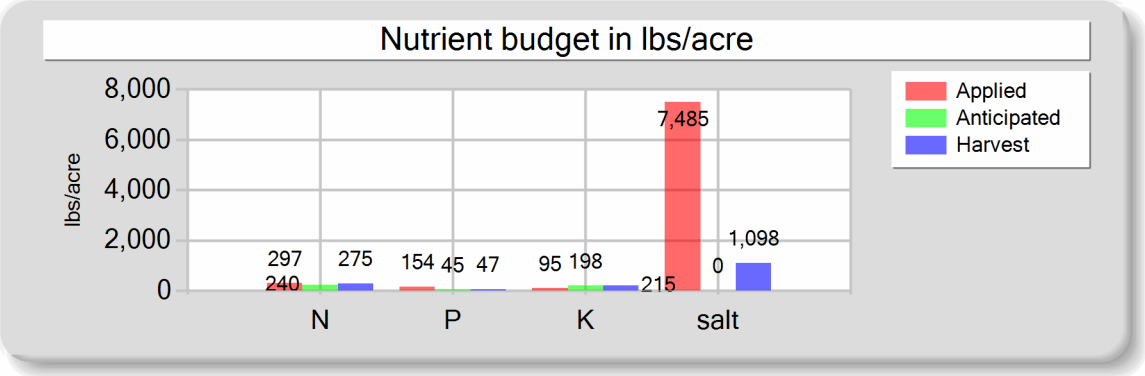
Fresh water applied	
28,713,600.00	*gallons*
1,057.42	*acre-inches*
11.25	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/12/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	284.27	154.15	94.76	7,354.13
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	6.01	0.00	0.00	130.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.28	154.15	94.76	7,484.93
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	275.33	47.18	215.29	1,097.91
Nutrient balance	21.95	106.98	-120.54	6,387.02
Applied to removed ratio	1.08	3.27	0.44	6.82

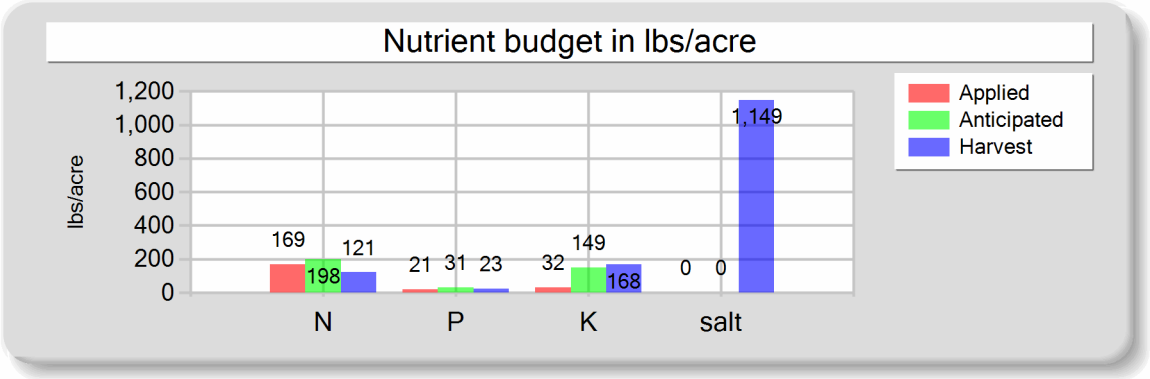
Fresh water applied	
128,219,724.00	*gallons*
4,721.90	*acre-inches*
50.23	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 10/27/2022: Wheat, silage, soft dough

Field name: 11 Crop: Wheat, silage, soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	161.80	20.95	32.47	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.53	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.33	20.95	32.47	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	121.27	23.32	167.91	1,148.53
Nutrient balance	48.07	-2.37	-135.44	-1,148.53
Applied to removed ratio	1.40	0.90	0.19	0.00

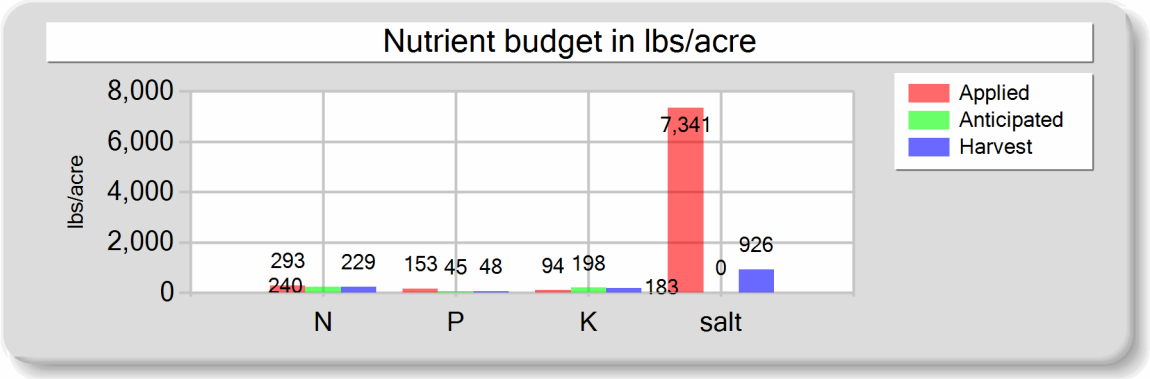
Fresh water applied	
10,440,000.00	*gallons*
384.47	*acre-inches*
11.65	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11 - 05/20/2023: Corn, silage

Field name: 11 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	281.64	152.73	93.88	7,286.30
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	3.91	0.00	0.00	54.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	292.55	152.73	93.88	7,340.68
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	229.48	48.31	183.18	925.95
Nutrient balance	63.08	104.42	-89.30	6,414.73
Applied to removed ratio	1.27	3.16	0.51	7.93

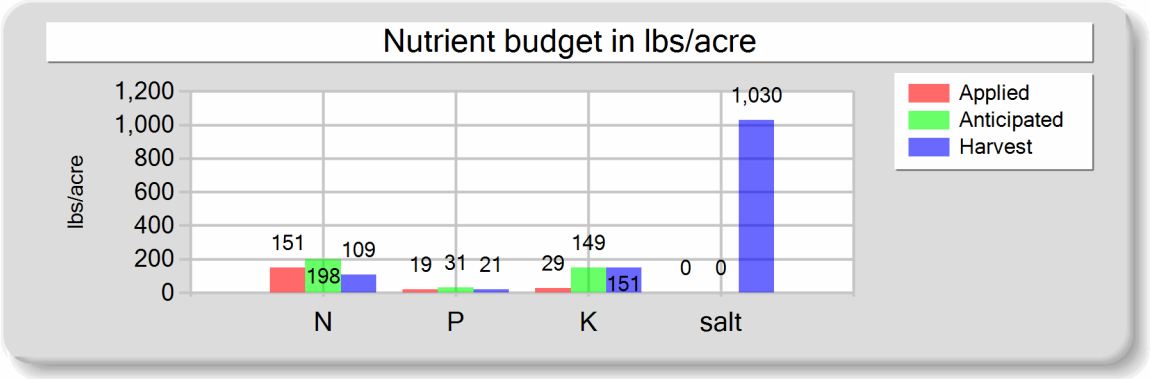
Fresh water applied	
46,560,000.00	*gallons*
1,714.65	*acre-inches*
51.96	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11a - 10/27/2022: Wheat, silage, soft dough

Field name: 11a | Crop: Wheat, silage, soft dough | Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	143.38	18.56	28.77	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.53	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	150.90	18.56	28.77	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	108.73	20.91	150.54	1,029.77
Nutrient balance	42.18	-2.35	-121.78	-1,029.77
Applied to removed ratio	1.39	0.89	0.19	0.00

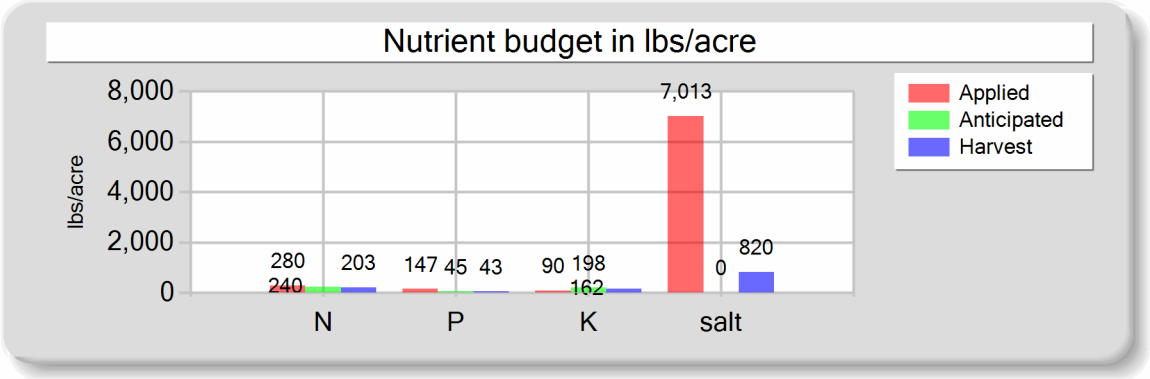
Fresh water applied	
2,844,687.00	*gallons*
104.76	*acre-inches*
11.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

11a - 05/26/2023: Corn, silage

Field name: 11a | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	271.08	147.00	90.36	7,013.07
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	2.35	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	280.43	147.00	90.36	7,013.07
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	203.17	42.77	162.18	819.81
Nutrient balance	77.26	104.23	-71.82	6,193.25
Applied to removed ratio	1.38	3.44	0.56	8.55

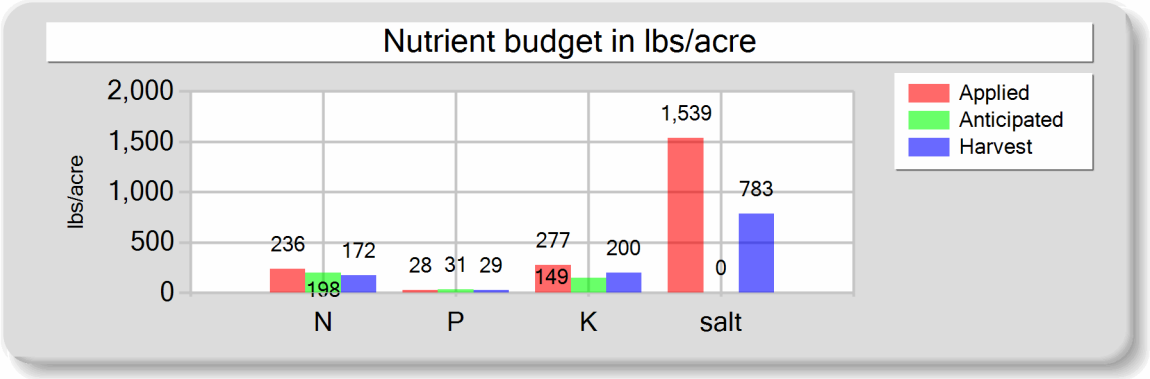
Fresh water applied	
12,686,649.00	*gallons*
467.21	*acre-inches*
51.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 11/02/2022: Wheat, silage, soft dough

Field name: 3 Crop: Wheat, silage, soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	227.10	27.69	276.86	1,474.20
Fresh water	2.33	0.00	0.00	65.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	236.43	27.69	276.86	1,539.40
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	171.65	28.80	200.45	783.36
Nutrient balance	64.78	-1.11	76.41	756.04
Applied to removed ratio	1.38	0.96	1.38	1.97

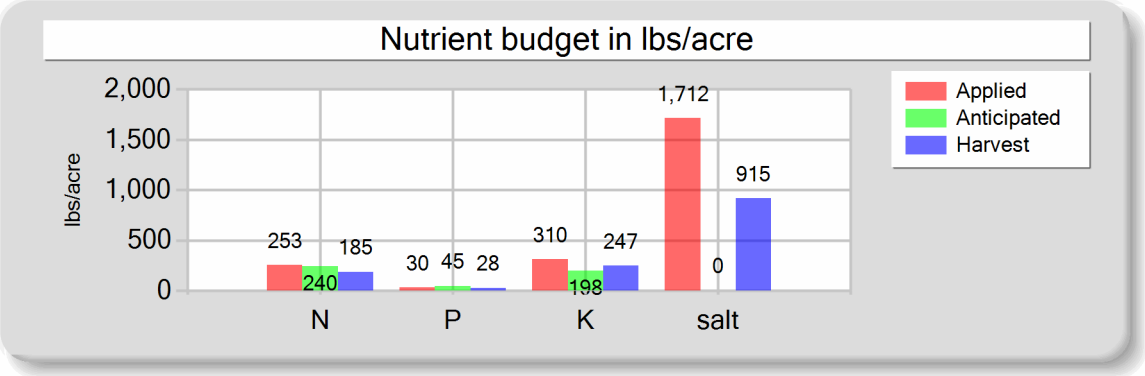
Fresh water applied	
5,175,289.00	*gallons*
190.59	*acre-inches*
13.61	*inches/acre*

Process wastewater applied	
576,000.00	*gallons*
21.21	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 05/30/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 05/30/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	240.40	30.30	309.69	1,556.75
Fresh water	5.56	0.00	0.00	155.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.96	30.30	309.69	1,712.48
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	184.53	28.05	246.53	915.24
Nutrient balance	68.43	2.25	63.16	797.24
Applied to removed ratio	1.37	1.08	1.26	1.87

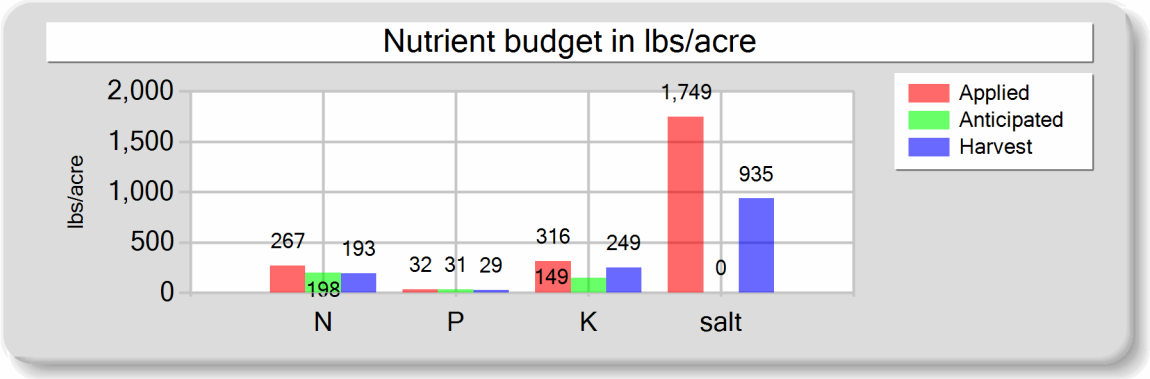
Fresh water applied	
20,627,144.00	*gallons*
759.63	*acre-inches*
54.26	*inches/acre*

Process wastewater applied	
768,000.00	*gallons*
28.28	*acre-inches*
2.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/18/2022: Wheat, silage, soft dough

Field name: 4 Crop: Wheat, silage, soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	257.94	31.59	315.68	1,683.65
Fresh water	2.33	0.00	0.00	65.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.27	31.59	315.68	1,748.84
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	192.72	28.76	248.81	934.85
Nutrient balance	74.54	2.83	66.87	813.99
Applied to removed ratio	1.39	1.10	1.27	1.87

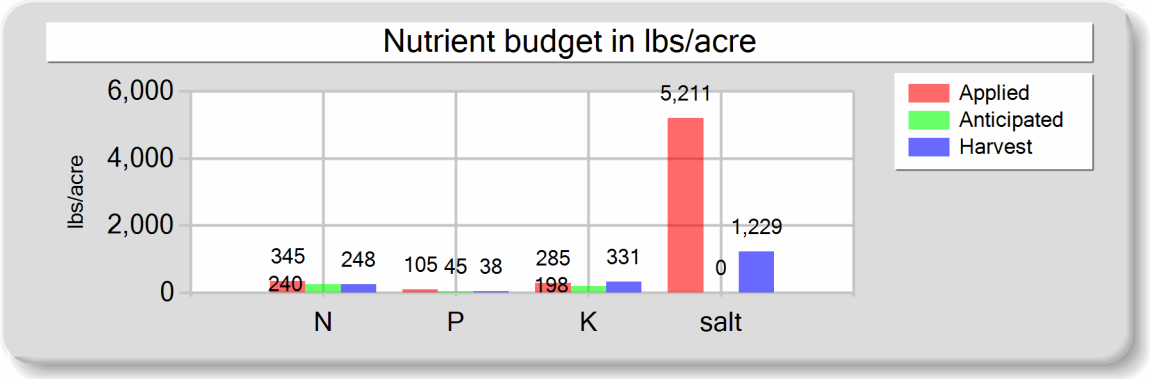
Fresh water applied	
13,623,152.00	*gallons*
501.69	*acre-inches*
13.56	*inches/acre*

Process wastewater applied	
1,728,000.00	*gallons*
63.64	*acre-inches*
1.72	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 05/26/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	150.72	81.73	50.24	3,899.16
Process wastewater	181.43	22.82	235.20	1,155.78
Fresh water	5.56	0.00	0.00	155.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	344.71	104.55	285.44	5,210.67
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	247.71	37.65	330.94	1,228.65
Nutrient balance	97.00	66.90	-45.50	3,982.02
Applied to removed ratio	1.39	2.78	0.86	4.24

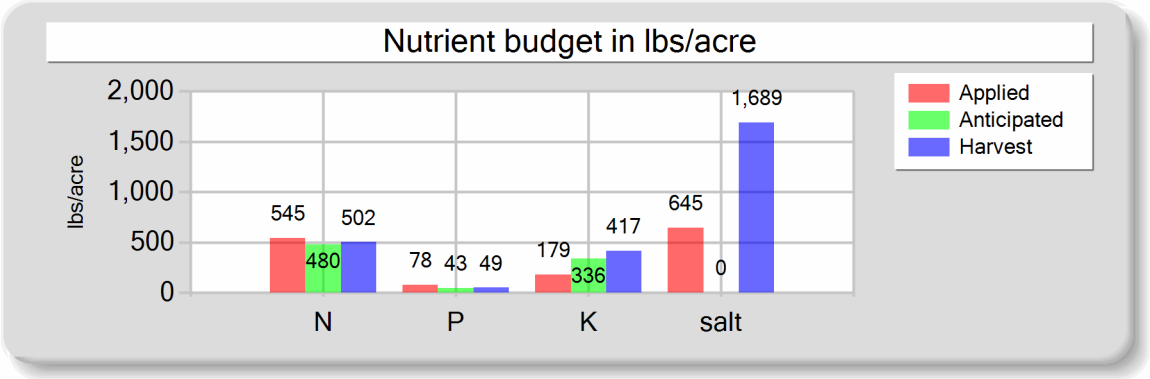
Fresh water applied	
54,030,470.00	*gallons*
1,989.76	*acre-inches*
53.78	*inches/acre*

Process wastewater applied	
1,632,000.00	*gallons*
60.10	*acre-inches*
1.62	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 11/01/2019: Alfalfa, hay

Field name: 5 Crop: Alfalfa, hay Plant date: 11/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	278.57	34.36	66.27	0.00
Process wastewater	241.21	43.70	112.68	341.42
Fresh water	10.83	0.00	0.00	303.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	544.61	78.06	178.95	644.65
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	501.57	48.71	416.58	1,689.47
Nutrient balance	43.04	29.35	-237.63	-1,044.82
Applied to removed ratio	1.09	1.60	0.43	0.38

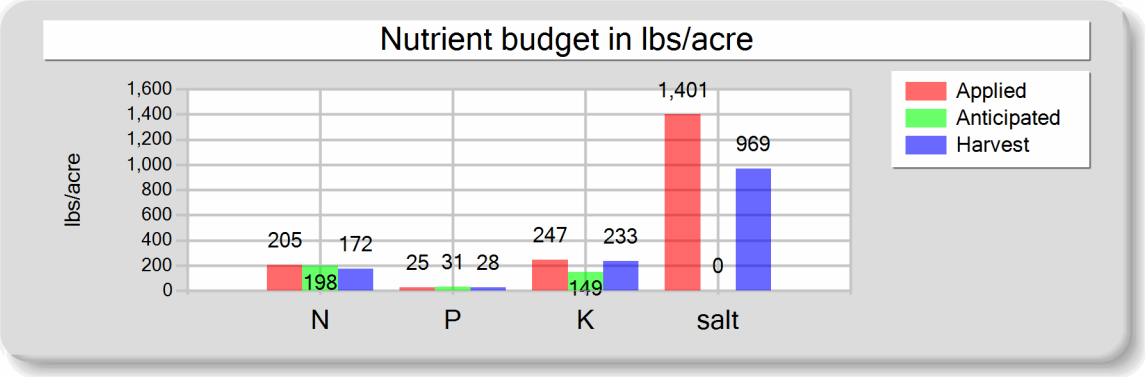
Fresh water applied	
71,375,359.00	*gallons*
2,628.51	*acre-inches*
47.79	*inches/acre*

Process wastewater applied	
768,000.00	*gallons*
28.28	*acre-inches*
0.51	*inches/acre*

Total harvests for the crop	
8	*harvests*

6 - 10/18/2022: Wheat, silage, soft dough

Field name: 6 Crop: Wheat, silage, soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.02	24.86	247.38	1,335.70
Fresh water	2.33	0.00	0.00	65.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	205.35	24.86	247.38	1,400.89
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	172.36	27.83	233.38	968.85
Nutrient balance	32.99	-2.98	14.00	432.04
Applied to removed ratio	1.19	0.89	1.06	1.45

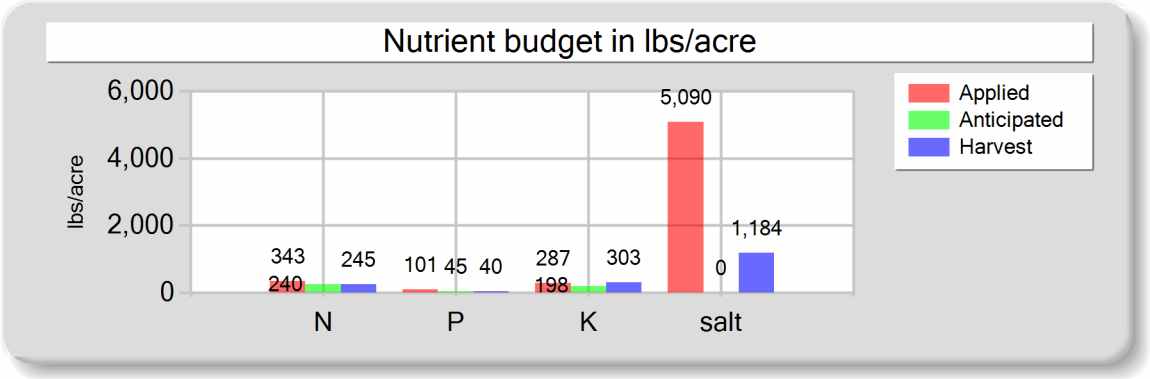
Fresh water applied	
26,688,296.00	*gallons*
982.84	*acre-inches*
13.65	*inches/acre*

Process wastewater applied	
2,544,000.00	*gallons*
93.69	*acre-inches*
1.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 05/20/2023: Corn, silage

Field name: 6 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	142.80	77.44	47.60	3,694.38
Process wastewater	187.61	23.70	239.79	1,239.46
Fresh water	5.56	0.00	0.00	155.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	342.97	101.14	287.39	5,089.57
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	245.35	40.10	302.63	1,183.78
Nutrient balance	97.63	61.04	-15.24	3,905.79
Applied to removed ratio	1.40	2.52	0.95	4.30

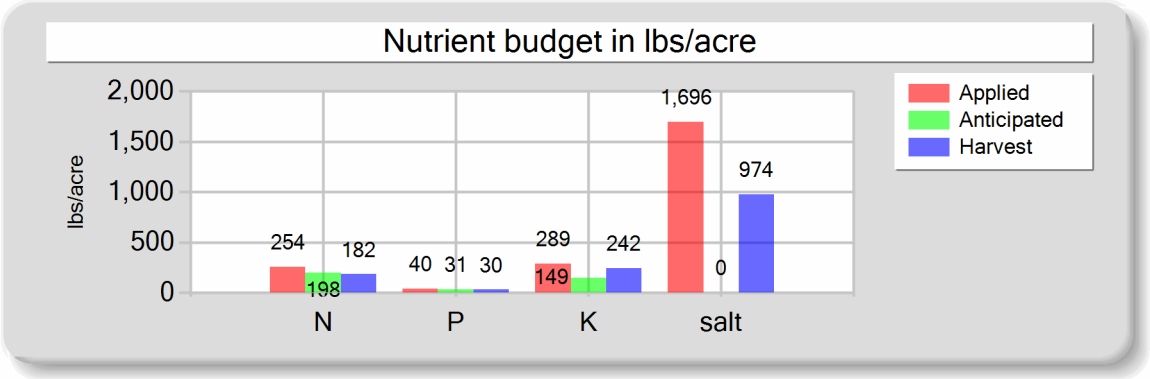
Fresh water applied	
106,727,943.00	*gallons*
3,930.43	*acre-inches*
54.59	*inches/acre*

Process wastewater applied	
2,832,000.00	*gallons*
104.29	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 10/19/2022: Wheat, silage, soft dough

Field name: 7 Crop: Wheat, silage, soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	239.20	39.88	289.27	1,630.92
Fresh water	7.38	0.00	0.00	65.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	253.57	39.88	289.27	1,696.11
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	182.49	29.90	242.29	974.33
Nutrient balance	71.08	9.98	46.98	721.78
Applied to removed ratio	1.39	1.33	1.19	1.74

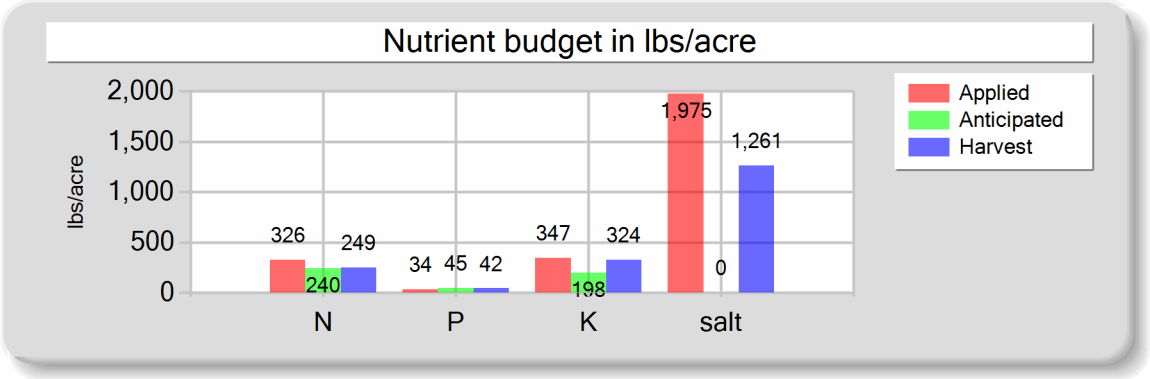
Fresh water applied	
28,125,309.00	*gallons*
1,035.76	*acre-inches*
13.81	*inches/acre*

Process wastewater applied	
3,696,000.00	*gallons*
136.11	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 05/29/2023: Corn, silage

Field name: 7 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	272.42	34.46	346.70	1,818.92
Fresh water	46.27	0.00	0.00	155.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.69	34.46	346.70	1,974.65
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	248.50	42.34	323.97	1,260.92
Nutrient balance	77.19	-7.88	22.72	713.73
Applied to removed ratio	1.31	0.81	1.07	1.57

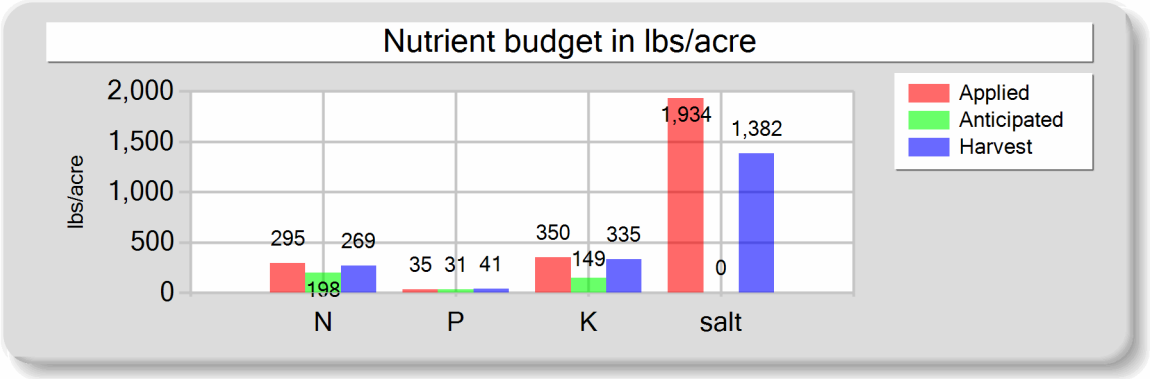
Fresh water applied	
108,067,441.00	*gallons*
3,979.76	*acre-inches*
53.06	*inches/acre*

Process wastewater applied	
4,080,000.00	*gallons*
150.25	*acre-inches*
2.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/17/2022: Wheat, silage, soft dough

Field name: 9 Crop: Wheat, silage, soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	285.45	35.05	350.13	1,869.03
Fresh water	2.33	0.00	0.00	65.19
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	294.77	35.05	350.13	1,934.22
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	268.82	41.24	334.50	1,382.30
Nutrient balance	25.95	-6.19	15.63	551.92
Applied to removed ratio	1.10	0.85	1.05	1.40

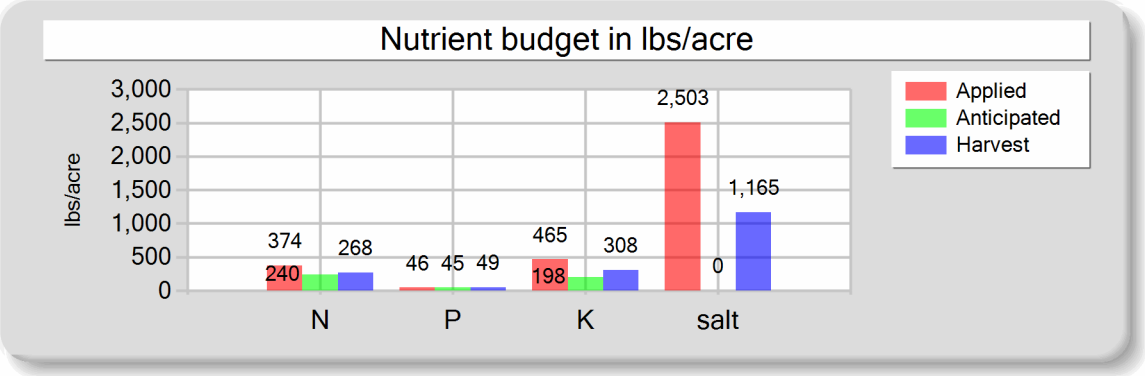
Fresh water applied	
26,688,296.00	*gallons*
982.84	*acre-inches*
13.65	*inches/acre*

Process wastewater applied	
3,720,000.00	*gallons*
136.99	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 05/20/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 05/20/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	361.32	45.55	464.86	2,347.73
Fresh water	5.56	0.00	0.00	155.73
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	373.88	45.55	464.86	2,503.46
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	267.69	49.34	307.58	1,165.25
Nutrient balance	106.19	-3.78	157.28	1,338.21
Applied to removed ratio	1.40	0.92	1.51	2.15

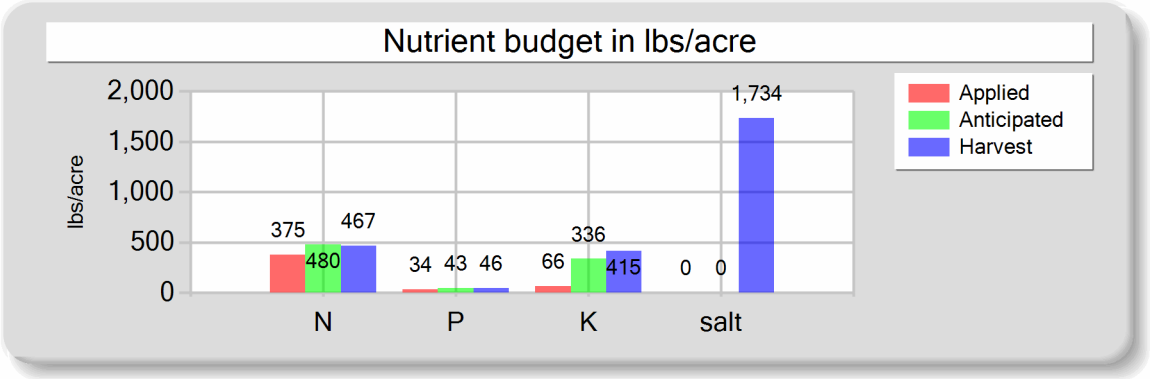
Fresh water applied	
106,727,943.00	*gallons*
3,930.43	*acre-inches*
54.59	*inches/acre*

Process wastewater applied	
5,856,000.00	*gallons*
215.66	*acre-inches*
3.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

B1 - 11/09/2020: Alfalfa, hay

Field name: B1 Crop: Alfalfa, hay Plant date: 11/09/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	278.85	34.40	66.33	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	81.96	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	374.81	34.40	66.33	0.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	467.28	45.84	415.02	1,733.53
Nutrient balance	-92.47	-11.45	-348.69	-1,733.53
Applied to removed ratio	0.80	0.75	0.16	0.00

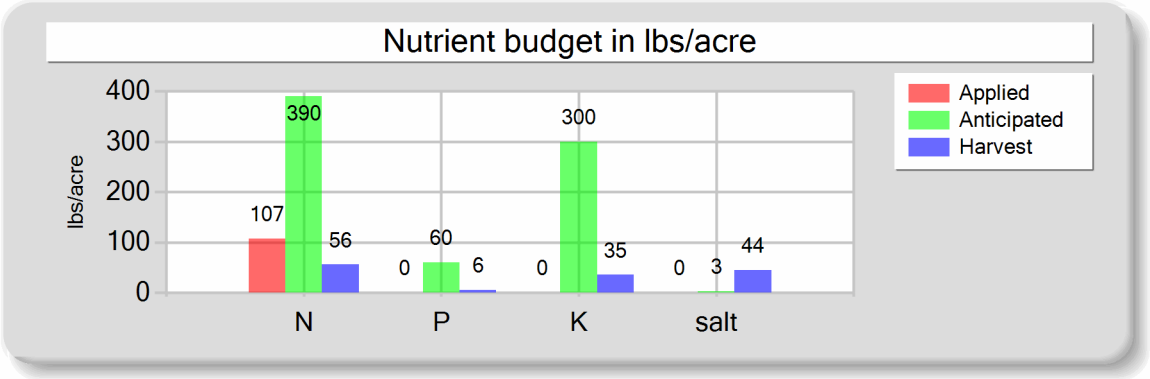
Fresh water applied	
151,200,000.00	*gallons*
5,568.18	*acre-inches*
47.59	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
8	*harvests*

B2 - 10/19/2022: Pistachios

Field name: B2 Crop: Pistachios Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	92.99	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	106.99	0.00	0.00	0.00
Anticipated crop nutrient removal	390.00	60.00	300.00	3.00
Actual crop nutrient removal	55.60	5.78	35.14	44.48
Nutrient balance	51.39	-5.78	-35.14	-44.48
Applied to removed ratio	1.92	0.00	0.00	0.00

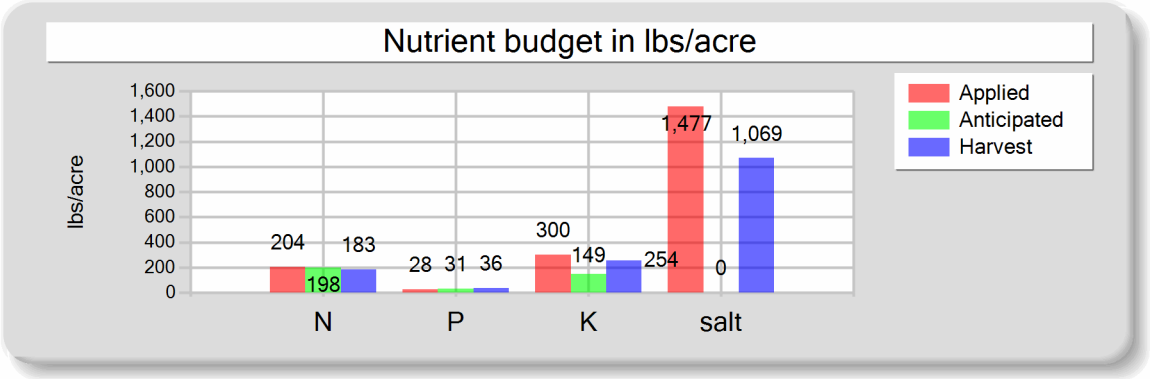
Fresh water applied	
50,000,000.00	*gallons*
1,841.33	*acre-inches*
52.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

G1 - 10/19/2022: Wheat, silage, soft dough

Field name: G1 Crop: Wheat, silage, soft dough Plant date: 10/19/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	182.15	27.76	300.26	1,477.24
Fresh water	15.22	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	204.37	27.76	300.26	1,477.24
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	182.74	35.62	253.98	1,068.57
Nutrient balance	21.63	-7.86	46.28	408.67
Applied to removed ratio	1.12	0.78	1.18	1.38

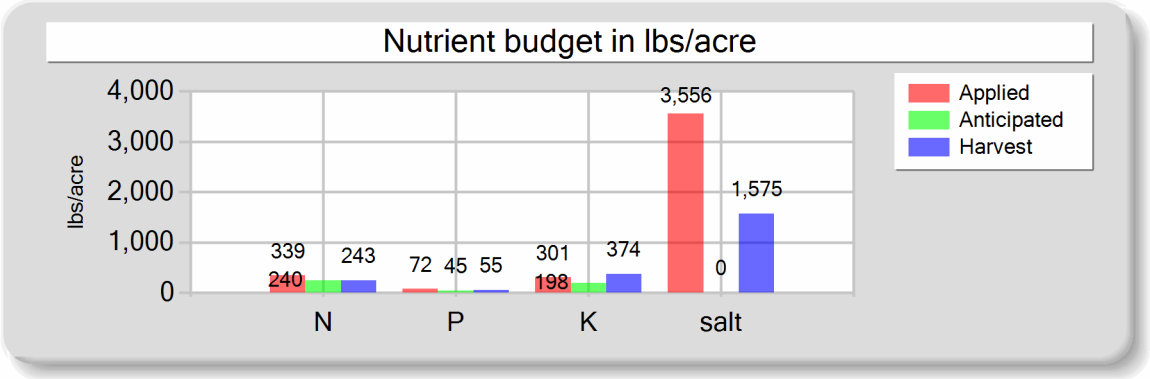
Fresh water applied	
9,840,000.00	*gallons*
362.37	*acre-inches*
10.66	*inches/acre*

Process wastewater applied	
3,360,000.00	*gallons*
123.74	*acre-inches*
3.64	*inches/acre*

Total harvests for the crop	
1	*harvests*

G1 - 05/25/2023: Corn, silage

Field name: G1 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	79.96	43.36	26.65	2,068.56
Process wastewater	208.81	28.89	274.60	1,356.96
Fresh water	43.64	0.00	0.00	130.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	339.41	72.25	301.25	3,556.32
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	242.57	55.32	374.49	1,574.58
Nutrient balance	96.84	16.92	-73.24	1,981.74
Applied to removed ratio	1.40	1.31	0.80	2.26

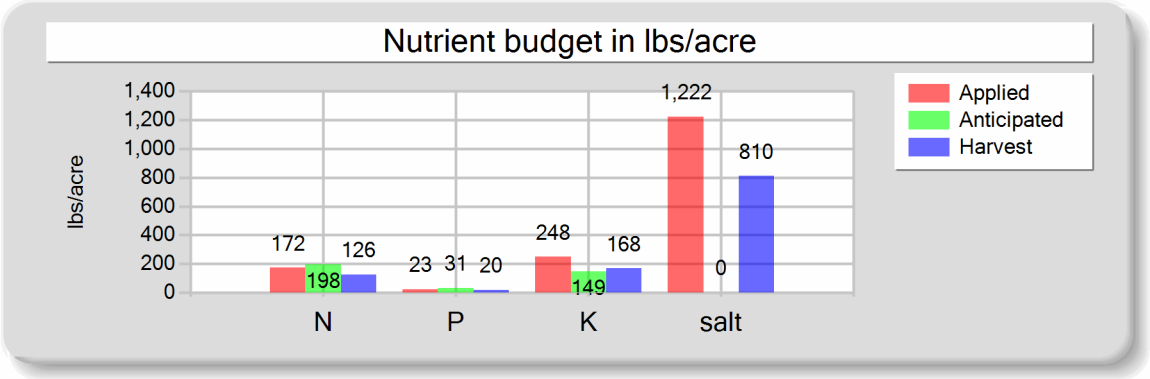
Fresh water applied	
44,232,666.00	*gallons*
1,628.94	*acre-inches*
47.91	*inches/acre*

Process wastewater applied	
3,048,000.00	*gallons*
112.25	*acre-inches*
3.30	*inches/acre*

Total harvests for the crop	
1	*harvests*

G2 - 10/20/2022: Wheat, silage, soft dough

Field name: G2 Crop: Wheat, silage, soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	150.41	22.97	248.42	1,221.96
Fresh water	14.98	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	172.38	22.97	248.42	1,221.96
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	125.66	20.03	167.55	810.44
Nutrient balance	46.72	2.94	80.87	411.53
Applied to removed ratio	1.37	1.15	1.48	1.51

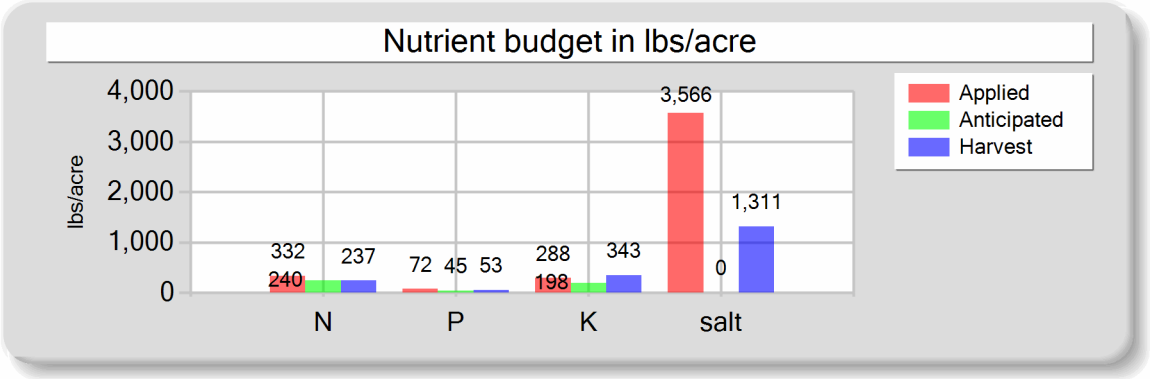
Fresh water applied	
10,824,000.00	*gallons*
398.61	*acre-inches*
10.49	*inches/acre*

Process wastewater applied	
3,120,000.00	*gallons*
114.90	*acre-inches*
3.02	*inches/acre*

Total harvests for the crop	
1	*harvests*

G2 - 05/25/2023: Corn, silage

Field name: G2 Crop: Corn, silage Plant date: 05/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	83.16	45.10	27.72	2,151.38
Process wastewater	197.65	27.29	259.84	1,283.83
Fresh water	43.72	0.00	0.00	130.80
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	331.53	72.38	287.56	3,566.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	236.81	52.86	342.53	1,310.92
Nutrient balance	94.72	19.52	-54.98	2,255.08
Applied to removed ratio	1.40	1.37	0.84	2.72

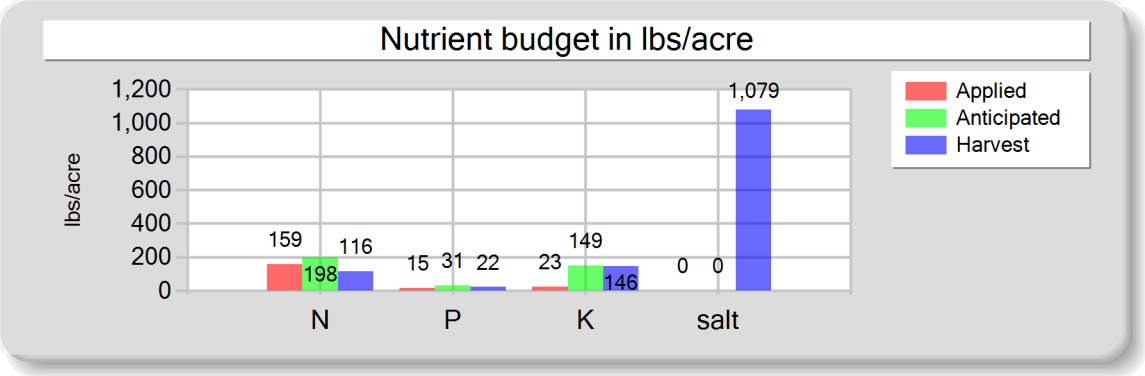
Fresh water applied	
49,495,803.00	*gallons*
1,822.76	*acre-inches*
47.97	*inches/acre*

Process wastewater applied	
3,192,000.00	*gallons*
117.55	*acre-inches*
3.09	*inches/acre*

Total harvests for the crop	
1	*harvests*

P2 - 10/22/2022: Wheat, silage, soft dough

Field name: P2 Crop: Wheat, silage, soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	114.42	14.81	22.96	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	37.30	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	158.72	14.81	22.96	0.00
Anticipated crop nutrient removal	198.00	30.60	149.40	0.00
Actual crop nutrient removal	115.70	21.59	145.92	1,079.33
Nutrient balance	43.02	-6.78	-122.97	-1,079.33
Applied to removed ratio	1.37	0.69	0.16	0.00

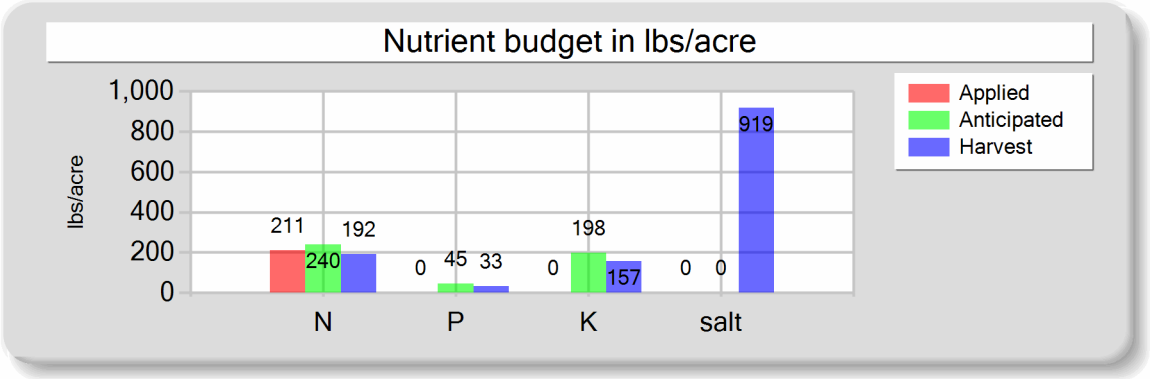
Fresh water applied	
19,080,000.00	*gallons*
702.65	*acre-inches*
10.04	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

P2 - 06/15/2023: Corn, silage

Field name: P2 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	203.56	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	210.56	0.00	0.00	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	191.51	33.00	157.22	918.72
Nutrient balance	19.05	-33.00	-157.22	-918.72
Applied to removed ratio	1.10	0.00	0.00	0.00

Fresh water applied	
141,300,000.00	*gallons*
5,203.60	*acre-inches*
54.77	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure - 2022

Sample and source description: Manure - 2022

Sample date: 10/10/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	30,900.00	4,000.00	6,200.00							0.00
DL	100.00	100.00	100.00							1.00

Corral Manure

Sample and source description: Corral Manure

Sample date: 10/04/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,100.00	10,900.00	6,700.00	29,600.00	6,100.00	1,100.00	7,200.00	0.10		52.00
DL	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		1.00

Separator Manure

Sample and source description: Separator Manure

Sample date: 12/11/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 76.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,700.00	2,800.00	5,400.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

Lagoon Q4 - 2022 - Dairy

Sample and source description: Lagoon Q4 - 2022 - Dairy

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	654.00	482.00	0.00	1.90	74.60	752.00								8,370.00	3,900
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q4 - 2022 - Heifers

Sample and source description: Lagoon Q4 - 2022 - Heifers

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.10

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	397.00	231.00	0.00	1.00	125.00	404.00								5,480.00	2,720
DL	0.01	0.01	0.01	0.01	0.01	0.01								0.01	1

Lagoon Q1 - Dairy

Sample and source description: Lagoon Q1 - Dairy

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	663.00	487.00	0.00	1.80	84.30	839.00								9,030.00	4,530
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q1 - Heifers

Sample and source description: Lagoon Q1 - Heifers

Sample date: 03/07/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	227.00	164.00	0.00	0.80	38.80	520.00								4,190.00	2,450
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2 - Dairy

Sample and source description: Lagoon Q2 - Dairy

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2,070.00	509.00	0.00	0.00	375.00	967.00								11,000.00	2,930
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2 - Heifers

Sample and source description: Lagoon Q2 - Heifers

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	40.30	13.60	0.00	0.10	12.50	63.40								630.00	333
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3 - Dairy

Sample and source description: Lagoon Q3 - Dairy

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.60

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	492.00	431.00	0.00	0.90	62.00	639.00								7,480.00	3,140
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3 - Heifers

Sample and source description: Lagoon Q3 - Heifers

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 8.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	71.90	59.90	0.00	0.60	23.90	173.00								1,870.00	992
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Slurry

Sample and source description: Slurry

Sample date: 09/27/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4,270.00	619.00	0.00	0.90	1,070.00	902.00								8,360.00	3,050
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4 - Dairy

Sample and source description: Lagoon Q4 - Dairy

Sample date: 12/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.50

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	684.00	464.00	0.00	1.50	93.10	758.00								8,790.00	3,700
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

MID Canal

Sample description: MID Canal

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value		0.00	1.00								32.20	28
DL		0.01	0.01								0.01	1

Domestic Well

Domestic Well

DW 1 & 2 Dairy

Sample description: DW 1 & 2 Dairy

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.40	0.00	7.40								513.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW Pombo North

Sample description: IW Pombo North

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.40	0.00	16.40								5,890.00	0
DL	0.01	0.01	0.01								10.00	1

IW Pombo South

Sample description: IW Pombo South

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	16.40	0.00	16.40								6,230.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW Pump 0

Sample description: IW Pump 0

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.20	0.00	0.20								1,310.00	0
DL	0.01	0.01	0.01								10.00	1

IW Pump 1

Sample description: IW Pump 1

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	21.60	0.00	21.60								936.00	0
DL	0.01	0.01	0.01								10.00	1

IW Pump 10

Sample description: IW Pump 10

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.30	0.00	6.30								999.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW Pump 11

Sample description: IW Pump 11

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								1,510.00	0
DL	0.01	0.01	0.01								10.00	1

IW Pump 29

Sample description: IW Pump 29

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.80	0.00	7.80								475.00	0
DL	0.01	0.01	0.01								10.00	1

IW Pump 31

Sample description: IW Pump 31

Sample date: 12/11/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.60	0.00	7.60								477.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

11

11

Field 11 Soil

Sample and source description: Field 11 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			33.00				
DL			0.01				

11a

Field 11A Soil

Sample and source description: Field 11A Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			33.00				
DL			0.01				

3

Field 3 Soil

Sample and source description: Field 3 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			32.00				
DL			0.01				

4

4

Field 4 Soil

Sample and source description: Field 4 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			35.00				
DL			0.01				

6

Field 6 Soil

Sample and source description: Field 6 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			32.00				
DL			0.01				

9

Field 9 Soil

Sample and source description: Field 9 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			0.01				

G1

G1

Field G1 Soil

Sample and source description: Field G1 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			38.00				
DL			0.01				

G2

Field G2 Soil

Sample and source description: Field G2 Soil

Sample date: 09/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			34.00				
DL			0.01				

E. PLANT TISSUE ANALYSES

10 - 10/26/2022: Wheat, silage, soft dough

Field 10 A Wheat

Sample and source description: Field 10 A Wheat

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,100.00	12,900.00		8.10
DL	100.00	10.00	100.00		1.00

10 - 10/26/2022: Wheat, silage, soft dough

Field 10 B Wheat

Sample and source description: Field 10 B Wheat

Sample date: 05/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,100.00	13,800.00		7.90
DL	100.00	100.00	100.00		1.00

10 - 06/12/2023: Corn, silage

Field 10A Corn

Sample and source description: Field 10A Corn

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	3,000.00	14,700.00		7.10
DL	0.01	0.01	0.01		1.00

Field 10B Corn

Sample and source description: Field 10B Corn

Sample date: 09/25/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,500.00	10,400.00		5.70
DL	0.01	0.01	0.01		1.00

11 - 10/27/2022: Wheat, silage, soft dough

11 - 10/27/2022: Wheat, silage, soft dough

Field 11 A Wheat

Sample and source description: Field 11 A Wheat

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,700.00	1,800.00	12,700.00		9.60
DL	100.00	100.00	100.00		1.00

Field 11 B Wheat

Sample and source description: Field 11 B Wheat

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,100.00	2,200.00	16,000.00		10.10
DL	100.00	100.00	100.00		1.00

11 - 05/20/2023: Corn, silage

Field 11 Corn Silage

Sample and source description: Field 11 Corn Silage

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,400.00	9,100.00		4.60
DL	0.01	0.01	0.01		1.00

3 - 11/02/2022: Wheat, silage, soft dough

3 - 11/02/2022: Wheat, silage, soft dough

Field 3 Wheat

Sample and source description: Field 3 Wheat

Sample date: 05/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 44.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,900.00	2,500.00	17,400.00		6.80
DL	100.00	100.00	100.00		1.00

4 - 10/18/2022: Wheat, silage, soft dough

Field 4 Wheat

Sample and source description: Field 4 Wheat

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 53.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,400.00	2,000.00	17,300.00		6.50
DL	100.00	100.00	100.00		1.00

4 - 05/26/2023: Corn, silage

Field 4 Corn Silage

Sample and source description: Field 4 Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	1,900.00	16,700.00		6.20
DL	0.01	0.01	0.01		1.00

5 - 11/01/2019: Alfalfa, hay

Field 5A & 5B Alfalfa Green Chop

Sample and source description: Field 5A & 5B Alfalfa Green Chop

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	39,700.00	3,700.00	29,700.00		12.90
DL	0.01	0.01	0.01		1.00

Field 5A-Alfalfa Haylage

Sample and source description: Field 5A-Alfalfa Haylage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,900.00	2,800.00	23,200.00		8.80
DL	0.01	0.01	0.01		1.00

Field 5B-Alfalfa Haylage

Sample and source description: Field 5B-Alfalfa Haylage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	26,100.00	2,800.00	31,600.00		10.30
DL	0.01	0.01	0.01		1.00

6 - 10/18/2022: Wheat, silage, soft dough

6 - 10/18/2022: Wheat, silage, soft dough

Field 6 A Wheat

Sample and source description: Field 6 A Wheat

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,700.00	2,900.00	22,700.00		9.20
DL	100.00	100.00	100.00		1.00

Field 6 B Wheat

Sample and source description: Field 6 B Wheat

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,300.00	20,900.00		8.90
DL	100.00	100.00	100.00		1.00

6 - 05/20/2023: Corn, silage

Field 6A Corn Silage

Sample and source description: Field 6A Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,500.00	2,000.00	15,700.00		6.10
DL	0.01	0.01	0.01		1.00

6 - 05/20/2023: Corn, silage

Field 6B Corn Silage

Sample and source description: Field 6B Corn Silage

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,200.00	2,200.00	16,000.00		6.30
DL	0.01	0.01	0.01		1.00

7 - 10/19/2022: Wheat, silage, soft dough

Field 7 A Wheat

Sample and source description: Field 7 A Wheat

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,800.00	2,900.00	24,800.00		10.60
DL	100.00	100.00	100.00		1.00

Field 7 B Wheat

Sample and source description: Field 7 B Wheat

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,500.00	2,900.00	22,100.00		8.30
DL	100.00	100.00	100.00		1.00

7 - 05/29/2023: Corn, silage

7 - 05/29/2023: Corn, silage

Field 7A Corn Silage

Sample and source description: Field 7A Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,300.00	2,400.00	20,400.00		7.30
DL	0.01	0.01	0.01		1.00

Field 7B Corn Silage

Sample and source description: Field 7B Corn Silage

Sample date: 09/11/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,200.00	14,800.00		6.40
DL	0.01	0.01	0.01		1.00

9 - 10/17/2022: Wheat, silage, soft dough

Field 9A Wheat

Sample and source description: Field 9A Wheat

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,300.00	2,500.00	22,500.00		8.60
DL	100.00	100.00	100.00		1.00

9 - 10/17/2022: Wheat, silage, soft dough

Field 9B Wheat

Sample and source description: Field 9B Wheat

Sample date: 04/30/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,900.00	2,800.00	21,200.00		9.50
DL	100.00	100.00	100.00		1.00

9 - 05/20/2023: Corn, silage

Field 9A Corn Silage

Sample and source description: Field 9A Corn Silage

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,600.00	2,100.00	13,000.00		5.70
DL	0.01	0.01	0.01		1.00

Field 9B Corn Silage

Sample and source description: Field 9B Corn Silage

Sample date: 09/09/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,900.00	2,600.00	16,300.00		5.40
DL	0.01	0.01	0.01		1.00

B1 - 11/09/2020: Alfalfa, hay

B1 - 11/09/2020: Alfalfa, hay

B1-A Alfalfa

Sample and source description: B1-A Alfalfa
Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 72.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	42,900.00	4,200.00	33,800.00		14.20
DL	0.01	0.01	0.01		1.00

B1-b Alfalfa

Sample and source description: B1-b Alfalfa
Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	42,300.00	4,500.00	34,600.00		14.50
DL	0.01	0.01	0.01		1.00

Field B1A-Alfalfa Haylage

Sample and source description: Field B1A-Alfalfa Haylage
Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight
Moisture: 37.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,900.00	2,500.00	29,300.00		12.20
DL	0.01	0.01	0.01		1.00

B1 - 11/09/2020: Alfalfa, hay

Field B1B-Alfalfa Haylage

Sample and source description: Field B1B-Alfalfa Haylage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,700.00	3,000.00	32,300.00		12.50
DL	0.01	0.01	0.01		1.00

G1 - 10/19/2022: Wheat, silage, soft dough

Field G1 Wheat

Sample and source description: Field G1 Wheat

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,800.00	2,300.00	16,400.00		6.90
DL	100.00	100.00	100.00		1.00

G1 - 05/25/2023: Corn, silage

Field G1 Corn Silage

Sample and source description: Field G1 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	2,600.00	17,600.00		7.40
DL	0.01	0.01	0.01		1.00

G2 - 10/20/2022: Wheat, silage, soft dough

Field G2 Wheat

Sample and source description: Field G2 Wheat

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,800.00	2,200.00	18,400.00		8.90
DL	100.00	100.00	100.00		1.00

G2 - 05/25/2023: Corn, silage

Field G2 Corn Silage

Sample and source description: Field G2 Corn Silage

Sample date: 09/08/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,200.00	2,500.00	16,200.00		6.20
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

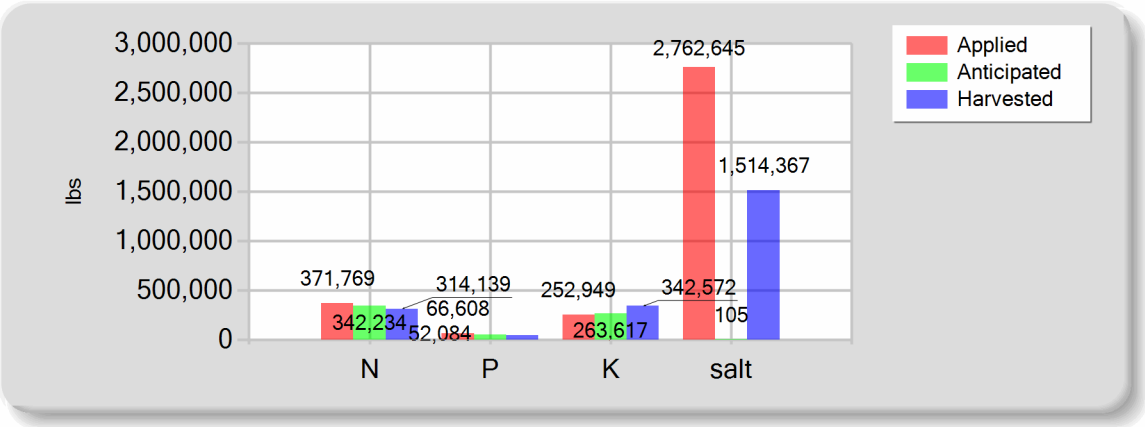
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

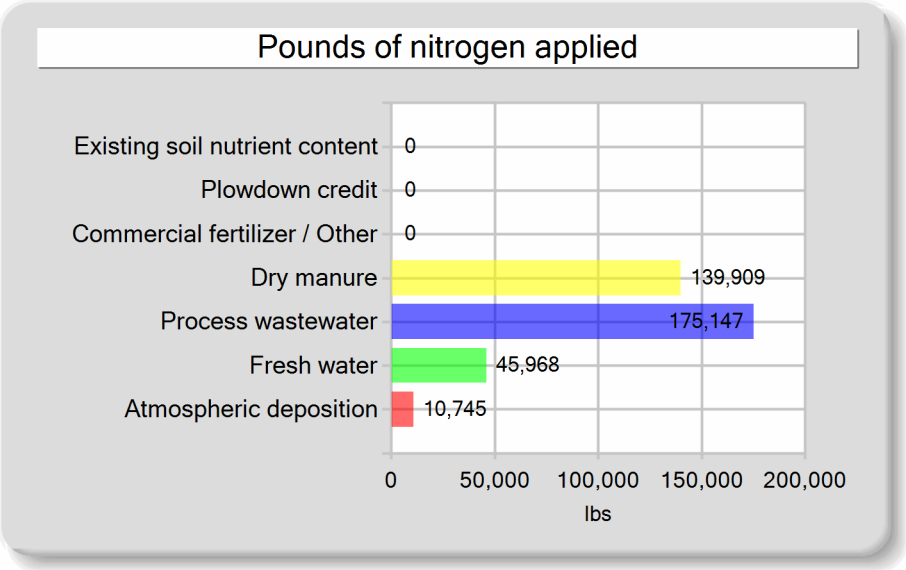
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

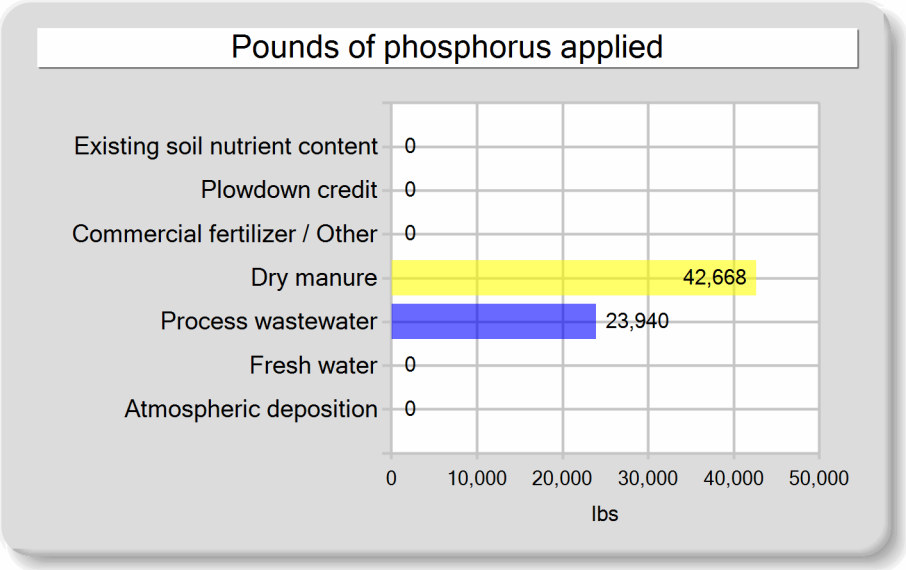
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	139,908.91	42,668.14	37,844.41	1,557,201.36
Process wastewater	175,146.72	23,940.03	215,104.16	1,105,608.49
Fresh water	45,968.17	0.00	0.00	99,834.84
Atmospheric deposition	10,745.00	0.00	0.00	0.00
Total nutrients applied	371,768.80	66,608.17	252,948.58	2,762,644.69
Anticipated crop nutrient removal	342,234.00	52,084.20	263,617.20	105.00
Actual crop nutrient removal	314,139.30	48,453.51	342,572.14	1,514,367.01
Nutrient balance	57,629.50	18,154.65	-89,623.56	1,248,277.68
Applied to removed ratio	1.18	1.37	0.74	1.82

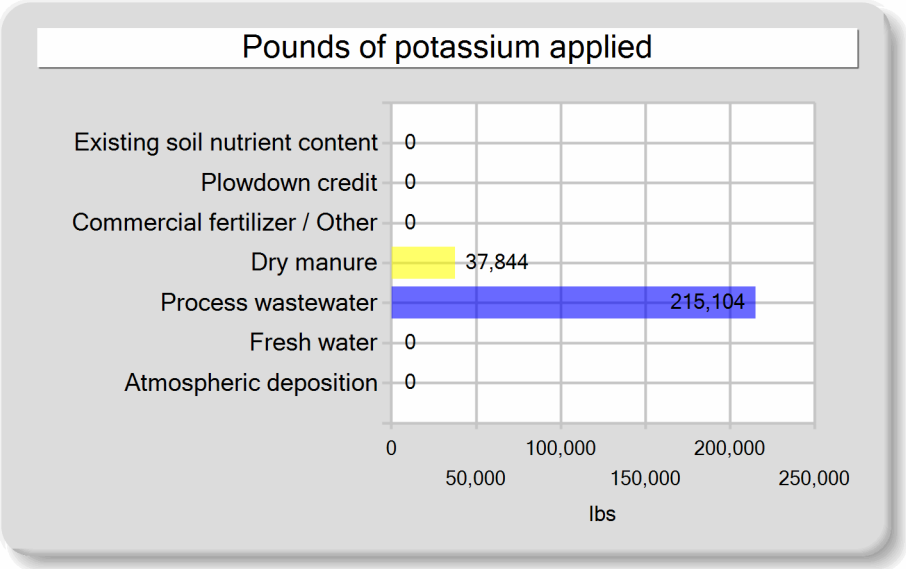
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

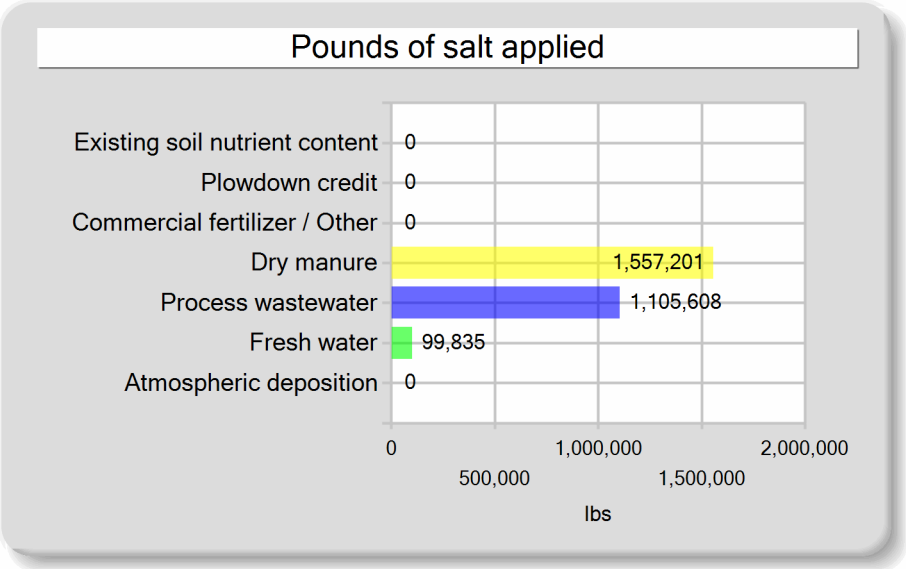


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Fields P1, P2, and P4 not planted therefore no application events submitted

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Paul Coelho	Paul Coelho
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Frank Coelho & Sons LP	Paul Coelho
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 1, 2024	Jul 1, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Frank Coelho & Sons Dairy LP
12775 Anchor Street
El Nido CA 95317
00-0021353 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0776
Sampled By: Trevor Coelho
Requested By: Paul Coelho / Mark Coelho
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	IW Pombo North	12/11/23	11:15	5890	5.89	16.4	ND		7.0
2	IW Pombo South	12/11/23	11:00	6230	6.23	16.4	ND		6.9
3	IW Pump 0	12/11/23	12:45	1310	1.31	0.2	ND		7.7
4	IW Pump 1	12/11/23	12:00	936	0.94	21.6	ND		7.5
5	IW Pump 10	12/11/23	13:15	999	1.00	6.3	ND		7.4
6	IW Pump 11	12/11/23	13:30	1510	1.51	ND	ND		8.0
7	IW Pump 31	12/11/23	16:00	477	0.48	7.6	ND		7.8
8	IW Pump 29	12/11/23	16:30	475	0.47	7.8	ND		7.8
9	DW 1 & 2 Dairy	12/11/23	17:00	513	0.51	7.4	ND		7.8

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Frank Coelho & Sons Dairy LP
12775 Anchor Street
El Nido CA 95317
00-0021353 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L0776
Sampled By: Trevor Coelho
Requested By: Paul Coelho / Mark Coelho
Submitted Date: 12/13/23
Reported Date: 12/20/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	IW Pombo North	12/11/23	11:15	**5890**	**5.89**	**16.4**	44.77	3.73
2	IW Pombo South	12/11/23	11:00	**6230**	**6.23**	**16.4**	44.77	3.73
3	IW Pump 0	12/11/23	12:45	**1310**	**1.31**	0.2	0.55	0.05
4	IW Pump 1	12/11/23	12:00	**936**	**0.94**	**21.6**	58.97	4.91
5	IW Pump 10	12/11/23	13:15	**999**	**1.00**	**6.3**	17.20	1.43
6	IW Pump 11	12/11/23	13:30	**1510**	**1.51**	ND	0.00	0.00
7	IW Pump 31	12/11/23	16:00	477	0.48	**7.6**	20.75	1.73
8	IW Pump 29	12/11/23	16:30	475	0.47	**7.8**	21.29	1.77
9	DW 1 & 2 Dairy	12/11/23	17:00	513	0.51	**7.4**	20.20	1.68

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.
ND = None Detected

DELLAVALLE™
LABORATORY INC

Frank Coelho & Sons Dairy LP
12775 Anchor Street
El Nido, CA 95317

Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L0776-01	IW Pombo North	Ag Water	Trevor Coelho		12/11/2023 11:15
23L0776-02	IW Pombo South	Ag Water	Trevor Coelho		12/11/2023 11:00
23L0776-03	IW Pump 0	Ag Water	Trevor Coelho		12/11/2023 12:45
23L0776-04	IW Pump 1	Ag Water	Trevor Coelho		12/11/2023 12:00
23L0776-05	IW Pump 10	Ag Water	Trevor Coelho		12/11/2023 13:15
23L0776-06	IW Pump 11	Ag Water	Trevor Coelho		12/11/2023 13:30
23L0776-07	IW Pump 31	Ag Water	Trevor Coelho		12/11/2023 16:00
23L0776-08	IW Pump 29	Ag Water	Trevor Coelho		12/11/2023 16:30
23L0776-09	DW 1 & 2 Dairy	Ag Water	Trevor Coelho		12/11/2023 17:00

Default Cooler Temperature on Receipt °C: 0.6
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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DELLAVALLE™
LABORATORY INC

Frank Coelho & Sons Dairy LP
12775 Anchor Street
El Nido, CA 95317

Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results

Sample: IW Pombo North
23L0776-01 (Water)

Sampled: 12/11/2023 11:15
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**5.89**	mmhos/cm	0.01	1		12/14/23 16:36	SM 2510 B		BEL0615
Electrical Conductivity umhos	**5890**	umhos/cm	10.0	1		12/14/23 16:36	SM 2510 B		BEL0615
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:15	Field		BEL0440
Nitrate Nitrogen as NO3N	**16.4**	mg/L	0.1	1	10	12/13/23 17:48	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:36	SM 4500-H+	H	BEL0615
pH	**7.0**	units	1.0	1		12/14/23 16:36	SM 4500-H+	H	BEL0615

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Account# 00-0021353
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Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: IW Pombo South
23L0776-02 (Water)

Sampled: 12/11/2023 11:00
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**6.23**	mmhos/cm	0.01	1		12/14/23 16:52	SM 2510 B		BEL0616
Electrical Conductivity umhos	**6230**	umhos/cm	10.0	1		12/14/23 16:52	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 11:00	Field		BEL0440
Nitrate Nitrogen as NO3N	**16.4**	mg/L	0.1	1	10	12/13/23 18:07	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:52	SM 4500-H+	H	BEL0616
pH	**6.9**	units	1.0	1		12/14/23 16:52	SM 4500-H+	H	BEL0616

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Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: IW Pump 0
23L0776-03 (Water)

Sampled: 12/11/2023 12:45
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.31**	mmhos/cm	0.01	1		12/14/23 16:54	SM 2510 B		BEL0616
Electrical Conductivity umhos	**1310**	umhos/cm	10.0	1		12/14/23 16:54	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 12:45	Field		BEL0440
Nitrate Nitrogen as NO3N	**0.2**	mg/L	0.1	1	10	12/13/23 18:27	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:54	SM 4500-H+	H	BEL0616
pH	**7.7**	units	1.0	1		12/14/23 16:54	SM 4500-H+	H	BEL0616

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Sample Results
(Continued)

Sample: IW Pump 1
23L0776-04 (Water)

Sampled: 12/11/2023 12:00
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.94**	mmhos/cm	0.01	1		12/14/23 16:56	SM 2510 B		BEL0616
Electrical Conductivity umhos	**936**	umhos/cm	10.0	1		12/14/23 16:56	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 12:00	Field		BEL0440
Nitrate Nitrogen as NO3N	**21.6**	mg/L	0.1	1	10	12/13/23 18:47	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:56	SM 4500-H+	H	BEL0616
pH	**7.5**	units	1.0	1		12/14/23 16:56	SM 4500-H+	H	BEL0616

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Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: IW Pump 10
23L0776-05 (Water)

Sampled: 12/11/2023 13:15
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.00**	mmhos/cm	0.01	1		12/14/23 16:57	SM 2510 B		BEL0616
Electrical Conductivity umhos	**999**	umhos/cm	10.0	1		12/14/23 16:57	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 13:15	Field		BEL0440
Nitrate Nitrogen as NO3N	**6.3**	mg/L	0.1	1	10	12/13/23 19:07	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:57	SM 4500-H+	H	BEL0616
pH	**7.4**	units	1.0	1		12/14/23 16:57	SM 4500-H+	H	BEL0616

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Received: 12/13/2023 7:00
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Sample Results
(Continued)

Sample: IW Pump 11
23L0776-06 (Water)

Sampled: 12/11/2023 13:30
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.51**	mmhos/cm	0.01	1		12/14/23 16:58	SM 2510 B		BEL0616
Electrical Conductivity umhos	**1510**	umhos/cm	10.0	1		12/14/23 16:58	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 13:30	Field		BEL0440
Nitrate Nitrogen as NO3N	ND	mg/L	0.1	1	10	12/13/23 19:26	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 16:58	SM 4500-H+	H	BEL0616
pH	**8.0**	units	1.0	1		12/14/23 16:58	SM 4500-H+	H	BEL0616

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Account# 00-0021353
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Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: IW Pump 31
23L0776-07 (Water)

Sampled: 12/11/2023 16:00
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.48**	mmhos/cm	0.01	1		12/14/23 17:00	SM 2510 B		BEL0616
Electrical Conductivity umhos	**477**	umhos/cm	10.0	1		12/14/23 17:00	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 16:00	Field		BEL0440
Nitrate Nitrogen as NO3N	**7.6**	mg/L	0.1	1	10	12/13/23 19:46	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 17:00	SM 4500-H+	H	BEL0616
pH	**7.8**	units	1.0	1		12/14/23 17:00	SM 4500-H+	H	BEL0616

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Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: IW Pump 29
23L0776-08 (Water)

Sampled: 12/11/2023 16:30
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.47**	mmhos/cm	0.01	1		12/14/23 17:01	SM 2510 B		BEL0616
Electrical Conductivity umhos	**475**	umhos/cm	10.0	1		12/14/23 17:01	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 16:30	Field		BEL0440
Nitrate Nitrogen as NO3N	**7.8**	mg/L	0.1	1	10	12/13/23 22:25	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 17:01	SM 4500-H+	H	BEL0616
pH	**7.8**	units	1.0	1		12/14/23 17:01	SM 4500-H+	H	BEL0616

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El Nido, CA 95317

Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Sample Results
(Continued)

Sample: DW 1 & 2 Dairy
23L0776-09 (Water)

Sampled: 12/11/2023 17:00
Sampled By: Trevor Coelho

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.51**	mmhos/cm	0.01	1		12/14/23 17:02	SM 2510 B		BEL0616
Electrical Conductivity umhos	**513**	umhos/cm	10.0	1		12/14/23 17:02	SM 2510 B		BEL0616
Ammonia (as N)	ND	mg/L	0.00	1		12/11/23 17:00	Field		BEL0440
Nitrate Nitrogen as NO3N	**7.4**	mg/L	0.1	1	10	12/13/23 22:45	EPA 300.0		BEL0446
Temperature	**25.0**	units	0.0	1		12/14/23 17:02	SM 4500-H+	H	BEL0616
pH	**7.8**	units	1.0	1		12/14/23 17:02	SM 4500-H+	H	BEL0616

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El Nido, CA 95317

Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0446										
Blank (BEL0446-BLK1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0446-BLK2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0446-BLK3)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0446-BLK4)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0446-BLK5)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0446-BS1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000		95.9	90-110		
LCS (BEL0446-BS2)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.9	90-110		
LCS (BEL0446-BS3)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.0		0.1	mg/L	5.000		99.7	90-110		
LCS (BEL0446-BS4)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.9		0.1	mg/L	5.000		97.9	90-110		
Duplicate (BEL0446-DUP1)		**Source: 23L0776-07**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	7.7		0.1	mg/L		7.6			1.53	10
Duplicate (BEL0446-DUP2)		**Source: 23L0778-01**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.6		0.1	mg/L		0.5			1.99	10
Duplicate (BEL0446-DUP3)		**Source: 23L0691-01**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.02		0.1	mg/L		0.02				10
Duplicate (BEL0446-DUP4)		**Source: 23L0774-09**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	0.04		0.1	mg/L		0.03			5.71	10
Matrix Spike (BEL0446-MS1)		**Source: 23L0776-07**			Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	12.7		0.1	mg/L	5.000	7.6	102	90-110		
Matrix Spike (BEL0446-MS2)		**Source: 23L0778-01**			Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L	5.000	0.5	95.6	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0446 (Continued)										
Matrix Spike (BEL0446-MS3)			**Source: 23L0691-01**		Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000	0.02	96.5	90-110		
Matrix Spike (BEL0446-MS4)			**Source: 23L0774-09**		Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	4.8		0.1	mg/L	5.000	0.03	95.8	90-110		
Reference (BEL0446-SRM1)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.9	90-110		
Reference (BEL0446-SRM2)					Prepared & Analyzed: 12/13/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		96.5	90-110		
Reference (BEL0446-SRM3)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.7			mg/L	10.00		96.5	90-110		
Reference (BEL0446-SRM4)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.8			mg/L	10.00		98.1	90-110		
Reference (BEL0446-SRM5)					Prepared & Analyzed: 12/14/2023					
Nitrate Nitrogen as NO3N	9.6			mg/L	10.00		95.6	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0615										
Blank (BEL0615-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	4.7		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0615-BLK2)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0615-BLK3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.5		1.0	units						
Duplicate (BEL0615-DUP1)		**Source: 23L0774-08**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.53		0.01	mmhos/cm		0.53			1.05	10
Electrical Conductivity umhos	528		10.0	umhos/cm		534			1.05	10
pH	8.3		1.0	units		8.3			0.00	10
Duplicate (BEL0615-DUP2)		**Source: 23L0776-01**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	5.91		0.01	mmhos/cm		5.89			0.203	10
pH	7.0		1.0	units		7.0			0.286	10
Electrical Conductivity umhos	5910		10.0	umhos/cm		5890			0.203	10
Reference (BEL0615-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	435			umhos/cm	426.0		102	90-110		
Reference (BEL0615-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0615-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0615-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0615-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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DELLAVALLE™ LABORATORY INC

Frank Coelho & Sons Dairy LP
12775 Anchor Street
El Nido, CA 95317

Account# 00-0021353
Account Manager: Ben Nydam
Submitted By: Paul Coelho / Mark Coelho

Received: 12/13/2023 7:00
Reported: 12/20/2023 14:19

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0615 (Continued)										
Reference (BEL0615-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0615-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0615-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0615-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0616										
Blank (BEL0616-BLK1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0616-BLK2)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	7.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0616-BLK3)					Prepared & Analyzed: 12/14/2023					
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.4		1.0	units						
Duplicate (BEL0616-DUP1)		**Source: 23L0777-02**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.90		0.01	mmhos/cm		0.90			0.0779	10
pH	7.5		1.0	units		7.5			0.00	10
Electrical Conductivity umhos	899		10.0	umhos/cm		898			0.0779	10
Duplicate (BEL0616-DUP2)		**Source: 23L0780-02**			Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	0.45		0.01	mmhos/cm		0.44			1.88	10
Electrical Conductivity umhos	446		10.0	umhos/cm		438			1.88	10
pH	7.5		1.0	units		7.4			0.402	10
Reference (BEL0616-SRM1)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	441			umhos/cm	426.0		103	90-110		
Reference (BEL0616-SRM2)					Prepared & Analyzed: 12/14/2023					
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0616-SRM3)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0616-SRM4)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0616-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Received: 12/13/2023 7:00
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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0616 (Continued)										
Reference (BEL0616-SRM5)					Prepared & Analyzed: 12/14/2023					
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0616-SRM6)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0616-SRM7)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		
Reference (BEL0616-SRM8)					Prepared & Analyzed: 12/14/2023					
pH	4.0			units	4.000		100	97.5-102.5		

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12/13/23 07:00 23L0776

JG

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

WATER WORK REQUEST

Bill To: Acct No. 21353 Cons. 8

Purchase Order No. _____ Results Needed By _____

Client: Coelho & Sons Dairy LP, Frank
Address: 12775 Anchor Street
City, State, Zip: El Nido, CA 95317
Phone _____ Fax _____
Cell/Email _____

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com
Requested by: Paul Coelho - (209) 617-6455
Mark Coelho - (209) 769-7911

Ranch _____

Date sampled: 12/11/23

Sampled by: Trevor Coelho

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

No. of Samples: 9 No. Bottles: 9

Water Type: [ ] Drinking [ ] Wastewater
[X] Ag Water [ ] Ground Water [ ] Mon. Well
[ ] Supply Water [ ] Other _____

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) 1 L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) 1 L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) 1 L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) 1 L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES	Sampled From:	Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C	#/#
1. IW Pombo North		12/11/23	1115A	Ø	0.6	1-2.4
2. IW Pombo South		12/11/23	11 Am	Ø	0.4	1-2.6
3. IW Pump 0		12/11/23	1245P	Ø	0.1	1-2.4
4. IW Pump 1		12/11/23	12pm	Ø	-0.3	1.8
5. IW Pump 10		12/11/23	115p	Ø	-0.1	1-2.1
6. IW Pump 11		12/11/23	130P	Ø	0.0	1.6
7. IW Pump 31		12/11/23	4pm	Ø	0.8	1-2.4
8. IW Pump 29		12/11/23	430P	Ø	-0.1	1-2.4
9. DW 1+2 Dairy		12/11/23	5pm	Ø	0.4	1-2.5
10.						

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)		Relinquished (Date/Time)	
First	TCoelho	Coelho Dairy	12/11/23	5pm	12/12/23	8AM
Second	Mpedroso	CASINC	12/12/23	8AM	12/12/23	3:30P
Third	UPS	DVL	12/12/23	3:36pm		
Fourth		DVL	12/13	07:00		

I guarantee that as the client, or on behalf of the client named above, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per m
If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative A n. them the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ____ Miles ____ Consulting ____

Amt Paid ____ Rec By ____ Check No. ____

Shipping
$ _____ In
$ _____ Out
Date _____

Signature _____

Sample received
[ ] Yes

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Laboratory

IR Thermometer SN: 221511276
Correction Factor: 0°C
Calibration Due: 03/06/2024
Location: Hanford

2020

Shipping Information: Shipped In ☐ Picked-Up ☐ Walk In ☑ DLI Sampler ☐ Other ☐

Container: Ice Chest ☑ Box ☐ None ☐ **Refrigerant:** Wet Ice ☑ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

12/13/23 07:00 23L0776

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

SENT OUT